CHAPTER

# 11

# Markov Chains

In general, the random variables within the family defining a stochastic process are not independent, and in fact can be statistically dependent in very complex ways. In this chapter we introduce the class of Markov random processes that have a simple form of dependence and that are quite useful in modeling many problems found in practice. We concentrate on integer-valued Markov processes, which are called Markov chains.

- Section 11.1 introduces Markov processes and the special case of Markov chains.
- Section 11.2 considers discrete-time Markov chains and examines the behavior of their state probabilities over time.
- Section 11.3 discusses structural properties of discrete-time Markov chains that determine their long-term behavior and limiting state probabilities.
- Section 11.4 introduces continuous-time Markov chains and considers the transient as well as long-term behavior of their state probabilities.
- Section 11.5 considers time-reversed Markov chains and develops interesting properties of reversible Markov chains that look the same going forwards and backwards in time.
- Finally, Section 11.6 introduces methods for simulating discrete-time and continuous-time Markov chains.

## 11.1 MARKOV PROCESSES

A random process $X(t)$ is a **Markov process** if the future of the process given the present is independent of the past, that is, if for arbitrary times $t_1 < t_2 < \cdots < t_k < t_{k+1}$,

$$\begin{aligned} P[X(t_{k+1}) = x_{k+1} \,|\, X(t_k) = x_k, \ldots, X(t_1) = x_1] \\ = P[X(t_{k+1}) = x_{k+1} \,|\, X(t_k) = x_k] \end{aligned} \tag{11.1}$$

if $X(t)$ is discrete-valued, and

$$\begin{aligned} P[a < X(t_{k+1}) \leq b \,|\, X(t_k) = x_k, \ldots, X(t_1) = x_1] \\ = P[a < X(t_{k+1}) \leq b \,|\, X(t_k) = x_k] \end{aligned} \tag{11.2a}$$

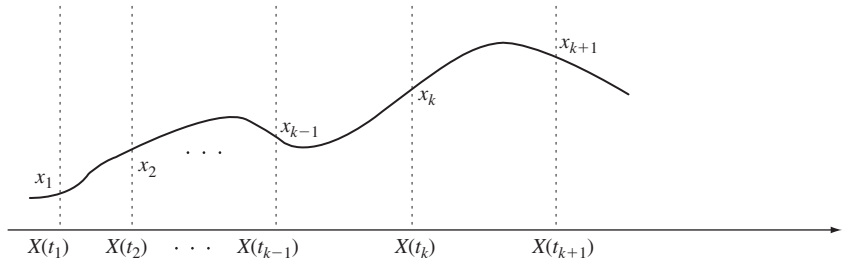



**FIGURE 11.1**
Markov property: Given $X(t_k)$, $X(t_{k+1})$ is independent of samples prior to $t_k$.

if $X(t)$ is continuous-valued. If the samples of $X(t)$ are jointly continuous, then Eq. (11.2a) is equivalent to

$$f_{X(t_{k+1})}(x_{k+1} \mid X(t_k) = x_k, \ldots, X(t_1) = x_1) = f_{X(t_{k+1})}(x_{k+1} \mid X(t_k) = x_k). \quad (11.2b)$$

We refer to Eqs. (11.1) and (11.2) as the **Markov property**. In the above expression $t_k$ is the "present," $t_{k+1}$ is the "future," and $t_1, \ldots, t_{k-1}$ is the "past," as shown in Fig. 11.1. Thus in Markov processes, pmf's and pdf's that are conditioned on several time instants always reduce to a pmf/pdf that is conditioned only on the most recent time instant. For this reason we refer to the value of $X(t)$ at time $t$ as the **state** of the process at time $t$.

### Example 11.1 Sum Process

Consider the sum process discussed in Section 9.3:

$$S_n = X_1 + X_2 + \cdots + X_n = S_{n-1} + X_n,$$

where the $X_i$'s are an iid sequence of random variables and where $S_0 = 0$. $S_n$ is a Markov process since

$$\begin{aligned} P[S_{n+1} = s_{n+1} \mid S_n = s_n, \ldots, S_1 = s_1] &= P[X_{n+1} = s_{n+1} - s_n] \\ &= P[S_{n+1} = s_{n+1} \mid S_n = s_n]. \end{aligned}$$

The binomial counting process and the random walk processes introduced in Section 9.3 are sum processes and therefore Markov processes.

### Example 11.2 Moving Average

Consider the moving average of a Bernoulli sequence:

$$Y_n = \frac{1}{2}(X_n + X_{n-1}),$$

where the $X_i$ are an independent Bernoulli sequence with $p = 1/2$. We now show that $Y_n$ is not a Markov process.

The pmf of $Y_n$ is

$$P[Y_n = 0] = P[X_n = 0, X_{n-1} = 0] = \frac{1}{4},$$

$$P\left[Y_n = \frac{1}{2}\right] = P[X_n = 0, X_{n-1} = 1] + P[X_n = 1, X_{n-1} = 0] = \frac{1}{2},$$

and

$$P[Y_n = 1] = P[X_n = 1, X_{n-1} = 1] = \frac{1}{4}.$$

Now consider the following conditional probability for two consecutive values of $Y_n$:

$$P\left[Y_n = 1 \,|\, Y_{n-1} = \frac{1}{2}\right] = \frac{P[Y_n = 1, Y_{n-1} = 1/2]}{P[Y_{n-1} = 1/2]}$$

$$= \frac{P[X_n = 1, X_{n-1} = 1, X_{n-2} = 0]}{1/2} = \frac{(1/2)^3}{1/2} = \frac{1}{4}.$$

Now suppose we have additional knowledge about the past:

$$P\left[Y_n = 1 \,|\, Y_{n-1} = \frac{1}{2}, Y_{n-2} = 1\right] = \frac{P[Y_n = 1, Y_{n-1} = 1/2, Y_{n-2} = 1]}{P[Y_{n-1} = 1/2, Y_{n-2} = 1]} = 0,$$

since no sequence of $X_n$'s leads to the sequence $1, 1/2, 1$. Thus

$$P\left[Y_n = 1 \,|\, Y_{n-1} = \frac{1}{2}, Y_{n-2} = 1\right] \neq P\left[Y_n = 1 \,|\, Y_{n-1} = \frac{1}{2}\right],$$

and the process is not Markov.

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### Example 11.3 Poisson Process

The Poisson process is a continuous-time Markov process since

$$P[N(t_{k+1}) = j \,|\, N(t_k) = i, N(t_{k-1}) = x_{k-1}, \ldots, N(t_1) = x_1]$$

$$= P[j - i \text{ events in } t_{k+1} - t_k \text{ seconds}]$$

$$= P[N(t_{k+1}) = j \,|\, N(t_k) = i].$$

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### Example 11.4 Random Telegraph

The random telegraph signal of Example 9.24 is a continuous-time Markov process since

$$P[X(t_{k+1}) = a \,|\, X(t_k) = b, \ldots, X(t_1) = x_1]$$

$$= P[\text{even (odd) number of jumps in } t_{k+1}$$

$$- t_k \text{ seconds if } a = b (a \neq b)]$$

$$= P[X(t_{k+1}) = a \,|\, X(t_k) = b].$$

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**Example 11.5 Wiener Process**

The Wiener process, from Section 9.5, is a Markov process. Since it satisfies the independent increments property (Eq. 9.52), we have that:

$$f_{X(t_{k+1})}(x_{k+1}|\, X(t_k) = x_k, \ldots, X(t_1) = x_1) = f_{X(t_{k+1}-t_k)}(x_{k+1} - x_k)$$
$$= \frac{\exp\left\{-\frac{1}{2}\left[\frac{(x_{k+1} - x_k)^2}{\alpha(t_{k+1} - t_k)}\right]\right\}}{\sqrt{(2\pi\alpha)(t_{k+1} - t_k)}}.$$

The Wiener process is Gaussian and so it provides an example of a Gaussian Markov process.

An integer-valued Markov random process is called a **Markov chain**.[1] In the remainder of this chapter we concentrate on Markov chains.

If $X(t)$ is a Markov chain, then the joint pmf for three arbitrary time instants is

$$\begin{aligned} P[X(t_3) = x_3, X(t_2) = x_2, X(t_1) = x_1] &= P[X(t_3) = x_3|X(t_2) = x_2, X(t_1) = x_1]P[X(t_2) = x_2, X(t_1) = x_1] \\ &= P[X(t_3) = x_3|X(t_2) = x_2]P[X(t_2) = x_2, X(t_1) = x_1] \\ &= P[X(t_3) = x_3|X(t_2) = x_2]P[X(t_2) = x_2|X(t_1) = x_1]P[X(t_1) = x_1], \end{aligned}$$

where we have used the definition of conditional probability and the Markov property. In general, the joint pmf for $k + 1$ arbitrary time instants is

$$\begin{aligned} P[X(t_{k+1}) = x_{k+1}, X(t_k) = x_k, \ldots, X(t_1) = x_1] &= P[X(t_{k+1}) = x_{k+1}|X(t_k) = x_k] \\ P[X(t_k) = x_k|X(t_{k-1}) = x_{k-1}] \ldots P[X(t_1) = x_1] &= \left\{\prod_{j=1}^{k} P[X(t_{j+1}) = x_{j+1}|X(t_j) = x_j]\right\} P[X(t_1) = x_1] \end{aligned} \tag{11.3}$$

Thus the *joint pmf of $X(t)$ at arbitrary time instants is given by the product of the pmf of the initial time instant and the probabilities for the subsequent state transitions*. Clearly, the state transition probabilities determine the statistical behavior of a Markov chain.

## 11.2 DISCRETE-TIME MARKOV CHAINS

Let $X_n$ be a discrete-time integer-valued Markov chain that starts at $n = 0$ with pmf

$$p_j(0) \triangleq P[X_0 = j] \qquad j = 0, 1, 2, \ldots. \tag{11.4}$$

[1]See Cox and Miller [6] for a discussion of continuous-valued Markov processes.

We will assume that $X_n$ takes on values from a countable set of integers, usually $\{0, 1, 2, \dots\}$. We say that the Markov chain is finite state if $X_n$ takes on values from a finite set.

From Eq. (11.3), the joint pmf for the first $n + 1$ values of the process is

$$\begin{aligned} &P[X_n = i_n, \dots, X_0 = i_0] \\ &\quad = P[X_n = i_n | X_{n-1} = i_{n-1}] \dots P[X_1 = i_1 | X_0 = i_0] P[X_0 = i_0]. \end{aligned} \tag{11.5}$$

Thus the joint pmf for a particular sequence is simply the product of the probability for the initial state and the probabilities for the subsequent one-step state transitions.

We will assume that the one-step state transition probabilities are fixed and do not change with time, that is,

$$P[X_{n+1} = j | X_n = i] = p_{ij} \qquad \text{for all } n. \tag{11.6}$$

$X_n$ is said to have **homogeneous transition probabilities**. The joint pmf for $X_n, \dots, X_0$ is then given by

$$P[X_n = i_n, \dots, X_0 = i_0] = p_{i_{n-1}, i_n} \dots p_{i_0, i_1} p_{i_0}(0). \tag{11.7}$$

Thus $X_n$ is completely specified by the *initial pmf* $p_i(0)$ and the *matrix of one-step transition probabilities* $P$:

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ . & . & . & \\ p_{i0} & p_{i1} & \cdots & \\ . & . & \cdots & \end{bmatrix}. \tag{11.8}$$

We will call $P$ the **transition probability matrix**. Note that each row of $P$ must add to one since

$$1 = \sum_j P[X_{n+1} = j | X_n = i] = \sum_j p_{ij}. \tag{11.9}$$

If the Markov chain is finite state, then the matrix $P$ will be an $n \times n$ nonnegative square with rows that add up to 1.

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### Example 11.6 Two-State Markov Chain for Speech Activity

A Markov model for packet speech assumes that if the $n$th packet contains silence, then the probability of silence in the next packet is $1 - \alpha$ and the probability of speech activity is $\alpha$. Similarly, if the $n$th packet contains speech activity, then the probability of speech activity in the next packet is $1 - \beta$ and the probability of silence is $\beta$.

Let $X_n$ be the indicator function for speech activity in a packet at time $n$, then $X_n$ is a two-state Markov chain with the state transition diagram shown in Fig. 11.2(a), and transition probability matrix

$$P = \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix}. \tag{11.10}$$

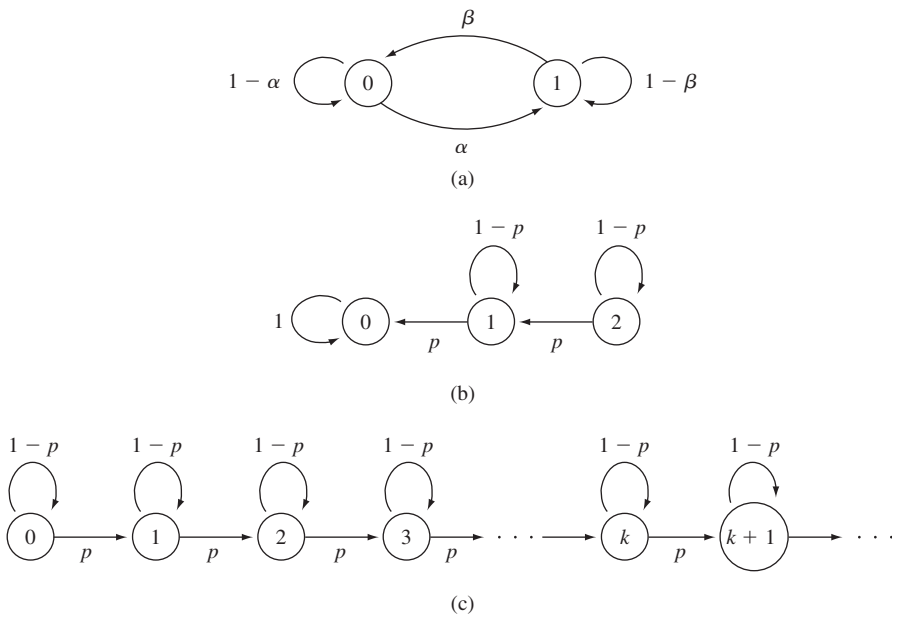



**FIGURE 11.2**
(a) State transition diagram for two-state Markov chain. (b) State transition diagram for Markov chain for light bulb inventory. (c) State transition diagram for binomial counting process.

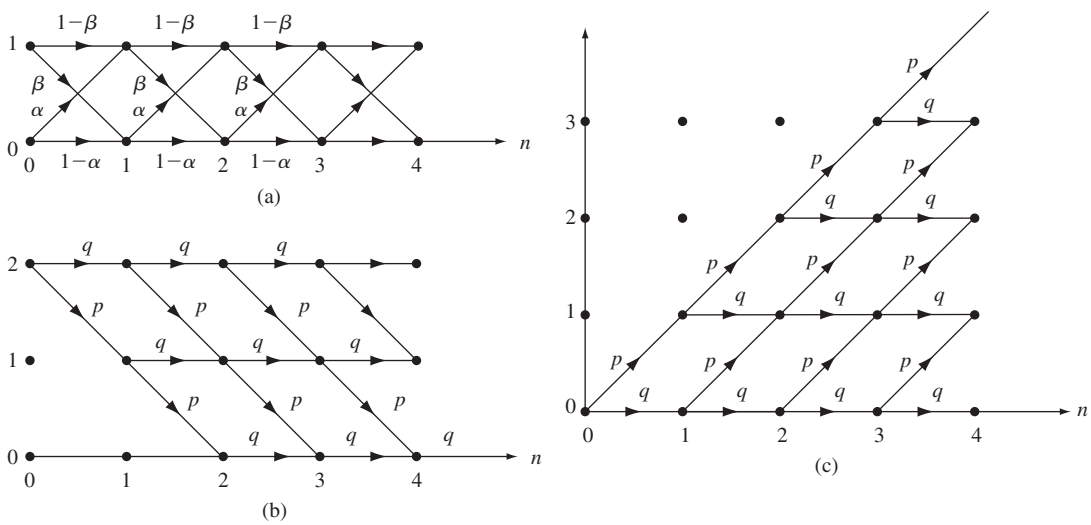



**FIGURE 11.3**
Trellis diagrams for Markov chain examples.

The sample functions of $X_n$ can be viewed as traversing the **trellis diagram** in Fig. 11.3(a) which shows the possible values of the process over time. At any give time, the process occupies the "state" that corresponds to its value. The sample function is realized as the process steps from one state at a given time instant to a state in the next time instant. The transitions are determined according to the transition probability matrix.

### Example 11.7

On day 0 a house has two new light bulbs in reserve. The probability that the house will need a single new light bulb during day $n$ is $p$, and the probability that it will not need any is $q = 1 - p$. Let $Y_n$ be the number of new light bulbs left in the house at the end of day $n$. $Y_n$ is a Markov chain with state transition diagram shown in Fig. 11.2(b), and transition probability matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ p & q & 0 \\ 0 & p & q \end{bmatrix}.$$

The trellis diagram for this process in Fig. 11.3(b) shows that, unless $q = 1$, the transition probabilities bias the process towards the "trapping" state $Y_n = 0$. Thus the sample functions of $Y_n$ are nonincreasing functions of $n$.

### Example 11.8 Binomial Counting Process

Let $S_n$ be the binomial counting process introduced in Example 9.15. In one step, $S_n$ can either stay the same or increase by one. The state transition diagram is shown in Fig. 11.2(c), and the transition probability matrix is given by

$$P = \begin{bmatrix} 1-p & p & 0 & 0 & \cdots \\ 0 & 1-p & p & 0 & \cdots \\ 0 & 0 & 1-p & p & \cdots \\ \cdot & \cdot & \cdots & & \cdots \end{bmatrix}.$$

The trellis diagram for binomial process in Fig. 11.3(c) shows that, unless $q = 1$, the transition probabilities bias the process towards steady growth over time. The sample functions of $S_n$ are nondecreasing functions of $n$.

### 11.2.1 The *n*-Step Transition Probabilities

To evaluate the joint pmf for arbitrary time instants (see Eq. 11.3), we need to know the transition probabilities for an arbitrary number of steps. Let $P(n) = \{p_{ij}(n)\}$ *be the matrix of n-step transition probabilities*, where

$$p_{ij}(n) = P[X_{n+k} = j \,|\, X_k = i] \qquad n \geq 0, i, j \geq 0. \tag{11.11}$$

Note that $P[X_{n+k} = j | X_k = i] = P[X_n = j | X_0 = i]$ for all $n \geq 0$ and $k \geq 0$, since the transition probabilities do not depend on time.

First, consider the two-step transition probabilities. The probability of going from state $i$ at $t = 0$, passing through state $k$ at $t = 1$, and ending at state $j$ at $t = 2$ is

$$\begin{aligned} P[X_2 = j, X_1 = k \,|\, X_0 = i] &= \frac{P[X_2 = j, X_1 = k, X_0 = i]}{P[X_0 = i]} \\ &= \frac{P[X_2 = j \,|\, X_1 = k]P[X_1 = k \,|\, X_0 = i]P[X_0 = i]}{P[X_0 = i]} \\ &= P[X_2 = j \,|\, X_1 = k]P[X_1 = k \,|\, X_0 = i] \\ &= p_{ik}(1)p_{kj}(1). \end{aligned}$$

Note that $p_{ik}(1)$ and $p_{kj}(1)$ are components of $P$, the one-step transition probability matrix. We obtain $p_{ij}(2)$, the probability of going from $i$ at $t = 0$ to $j$ at $t = 2$, by summing over all possible intermediate states $k$:

$$p_{ij}(2) = \sum_k p_{ik}(1)p_{kj}(1) \qquad \text{for all } i, j. \tag{11.12a}$$

Equation (11.12a) states that the $ij$ entry of $P(2)$ is obtained by multiplying the $i$th row of $P(1)$ by the $j$th column of $P(1)$. In other words, $P(2)$ is obtained by multiplying the one-step transition probability matrices:

$$P(2) = P(1)P(1) = P^2. \tag{11.12b}$$

Now consider the probability of going from state $i$ at $t = 0$, passing through state $k$ at $t = m$, and ending at state $j$ at time $t = m + n$. Following the same procedure as above we obtain the **Chapman–Kolmogorov equations**:

$$p_{ij}(m + n) = \sum_k p_{ik}(m)p_{kj}(n) \quad \text{for all } n, m \geq 0 \text{ all } i, j. \tag{11.13a}$$

Therefore the matrix of $n + m$ step transition probabilities $P(n + m) = \{p_{ij}(n + m)\}$ is obtained by the following matrix multiplication:

$$P(n + m) = P(n)P(m). \tag{11.13b}$$

It is easy to show by an induction argument that this implies that:

$$P(n) = P^n. \tag{11.14}$$

When the Markov chain has finite state, we can use computer programs to calculate the powers of $P$ numerically.

### 11.2.2 The State Probabilities

Now consider the state probabilities at time $n$. Let $\mathbf{p}(n) = \{p_j(n)\}$ denote the row vector of **state probabilities** at time $n$. The probability $p_j(n)$ is related to $\mathbf{p}(n - 1)$ by

$$\begin{aligned} p_j(n) &= \sum_i P[X_n = j \,|\, X_{n-1} = i]P[X_{n-1} = i] \\ &= \sum_i p_{ij}p_i(n - 1). \end{aligned} \tag{11.15a}$$

Equation (11.15a) states that $\mathbf{p}(n)$ is obtained by multiplying the row vector $\mathbf{p}(n-1)$ by the matrix $P$:

$$\mathbf{p}(n) = \mathbf{p}(n-1)P. \tag{11.15b}$$

Similarly, $p_j(n)$ is related to $\mathbf{p}(0)$ by

$$\begin{aligned} p_j(n) &= \sum_i P[X_n = j | X_0 = i]P[X_0 = i] \\ &= \sum_i p_{ij}(n)p_i(0), \end{aligned} \tag{11.16a}$$

and in matrix notation

$$\mathbf{p}(n) = \mathbf{p}(0)P(n) = \mathbf{p}(0)P^n \qquad n = 1, 2, \ldots. \tag{11.16b}$$

Thus the *state pmf at time n is obtained by multiplying the initial state pmf by* $P^n$.

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**Example 11.9**

To find the $n$-step transition probability in Example 11.7, note that

$$\begin{aligned} p_{22}(n) &= P[\text{no new light bulbs needed in } n \text{ days}] = q^n \\ p_{21}(n) &= P[1 \text{ light bulb needed in } n \text{ days}] = npq^{n-1} \\ p_{20}(n) &= 1 - p_{22}(n) - p_{21}(n). \end{aligned}$$

The other terms in $P(n)$ are found in similar fashion, thus

$$P(n) = \begin{bmatrix} 1 & 0 & 0 \\ 1 - q^n & q^n & 0 \\ 1 - q^n - npq^{n-1} & npq^{n-1} & q^n \end{bmatrix}.$$

Note that if $q < 1$ then, as $n \to \infty$,

$$P(n) \to \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

As a result, the state pmf $\mathbf{p}(n) = (p_0(n), p_1(n), p_2(n))$ approaches

$$\begin{aligned} \mathbf{p}(n) &= (p_0(0), p_1(0), p_2(0))P(n) \\ &= (0, 0, 1)P(n) \\ &\to (0, 0, 1)\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} = (1, 0, 0), \end{aligned}$$

where $(p_0(0), p_1(0), p_2(0))$ is the row vector of initial state probabilities and $(p_0(0), p_1(0), p_2(0)) = (0, 0, 1)$ since we start with two light bulbs. As time progresses, $p_0(n) \to 1$. In words, the above equation states that we eventually run out of light bulbs!

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### Example 11.10

Let $\alpha = 1/10$ and $\beta = 1/5$ in Example 11.6. Find $P(n)$ for $n = 2, 4, 8$, and 16.

$$P^2 = \begin{bmatrix} .9 & .1 \\ .2 & .8 \end{bmatrix}^2 = \begin{bmatrix} .83 & .17 \\ .34 & .66 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} .83 & .17 \\ .34 & .66 \end{bmatrix}^2 = \begin{bmatrix} .7467 & .2533 \\ .5066 & .4934 \end{bmatrix}$$

and similarly

$$P^8 = \begin{bmatrix} .6859 & .3141 \\ .6282 & .3718 \end{bmatrix} \qquad P^{16} = \begin{bmatrix} .6678 & .3322 \\ .6644 & .3356 \end{bmatrix}.$$

There is a clear trend here: It appears that as $n \to \infty$,

$$P^n \to \begin{bmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix}.$$

We can use matrix diagonalization methods from linear algebra to find $P^n$ [Anton, p. 246]. First we find that the eigenvalues of $P$ are 1 and $1 - \alpha - \beta$ from:

$$0 = \det(P - \lambda I) = \begin{vmatrix} 1 - \alpha - \lambda & \alpha \\ \beta & 1 - \beta - \lambda \end{vmatrix} = (1 - \alpha - \lambda)(1 - \beta - \lambda) - \alpha\beta$$
$$= (1 - \lambda)(1 - \alpha - \beta - \lambda).$$

The corresponding eigenvectors are:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{e}_2 = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

so the matrix with eigenvectors as columns is:

$$\mathbf{E} = [\mathbf{e}_1\, \mathbf{e}_2] = \begin{bmatrix} 1 & \alpha \\ 1 & -\beta \end{bmatrix}.$$

We then have that:

$$P = \mathbf{E}\Lambda\mathbf{E}^{-1} = \frac{1}{\alpha + \beta}\begin{bmatrix} 1 & \alpha \\ 1 & -\beta \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 - \alpha - \beta \end{bmatrix}\begin{bmatrix} \beta & \alpha \\ 1 & -1 \end{bmatrix}.$$

The payoff is in the calculation of $P^n$:

$$P^n = (\mathbf{E}\Lambda\mathbf{E}^{-1})(\mathbf{E}\Lambda\mathbf{E}^{-1})\ldots(\mathbf{E}\Lambda\mathbf{E}^{-1}) = \mathbf{E}\Lambda(\mathbf{E}^{-1}\mathbf{E})\Lambda\ldots\Lambda(\mathbf{E}^{-1}\mathbf{E})\Lambda\mathbf{E}^{-1}$$
$$= \mathbf{E}\Lambda\Lambda\ldots\Lambda\mathbf{E}^{-1} = \mathbf{E}\Lambda^n\mathbf{E}^{-1}$$
$$= \frac{1}{\alpha + \beta}\begin{bmatrix} 1 & \alpha \\ 1 & -\beta \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & (1 - \alpha - \beta)^n \end{bmatrix}\begin{bmatrix} \beta & \alpha \\ 1 & -1 \end{bmatrix}$$
$$= \begin{bmatrix} \dfrac{\beta}{\alpha + \beta} & \dfrac{\alpha}{\alpha + \beta} \\ \dfrac{\beta}{\alpha + \beta} & \dfrac{\alpha}{\alpha + \beta} \end{bmatrix} + \frac{(1 - \alpha - \beta)^n}{\alpha + \beta}\begin{bmatrix} \alpha & \alpha \\ -\beta & \beta \end{bmatrix}.$$

As long as $|1 - \alpha - \beta| < 1$, the second term goes to zero as $n \to \infty$ and so

$$P^n \to \begin{bmatrix} \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \\ \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}.$$

Note that all the rows are the same in the limiting matrix.

### Example 11.11

Let the initial state probabilities in Example 11.10 be

$$P[X_0 = 0] = p_0(0) \quad \text{and} \quad P[X_0 = 1] = 1 - p_0(0).$$

Find the state probabilities as $n \to \infty$.

The state probability vector at time $n$ is:

$$\begin{aligned} \mathbf{p}(n) &= (p_0(0), 1 - p_0(0)P^n \\ &= (p_0(0), 1 - p_0(0)\begin{bmatrix} \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \\ \frac{\beta}{\alpha+\beta} & \frac{\alpha}{\alpha+\beta} \end{bmatrix} + \frac{(1-\alpha-\beta)^n}{\alpha+\beta}(p_0(0), 1 - p_0(0)\begin{bmatrix} \alpha & \alpha \\ -\beta & \beta \end{bmatrix}. \end{aligned}$$

As $n \to \infty$, we have that

$$\mathbf{p}(n) \to (p_0(0), 1 - p_0(0))\begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{2}{3}, \frac{1}{3} \end{bmatrix}.$$

We see that the state probabilities do not depend on the initial state probabilities as $n \to \infty$.

### Example 11.12 Google PageRank

A Web surfer browses pages in a five-page Web universe shown in Fig. 11.4(a). The surfer selects the next page to view by selecting with equal probability from the pages pointed to by the current page. If a page has no outgoing link (e.g., page 2), then the surfer selects any of the pages in the universe with equal probability. Find the probability that the surfer views page $i$.

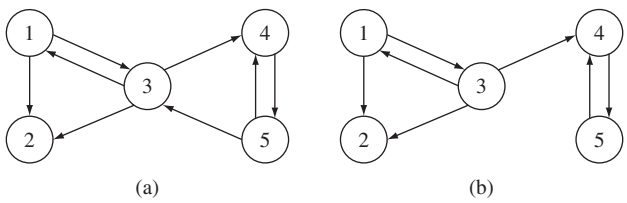



**FIGURE 11.4**
State-transition diagrams for PageRank examples.

The viewing behavior can be modeled by a Markov chain where the state represents the page currently viewed. If the current page points to $k$ pages, then the next page is selected from that group with probability $1/k$. If the current page does not point to any pages, then the next page can be any of the 5 pages with probability 1/5. The transition probability for the Markov chain is:

$$P = [p_{ij}] = \begin{bmatrix} 0 & 1/2 & 1/2 & 0 & 0 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/3 & 1/3 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1/2 & 0 & 1/2 \end{bmatrix}.$$

We can obtain the limiting state probabilities numerically by letting Octave calculate a high power of $P$, say $P^{50}$. We then obtain a $5 \times 5$ matrix in which all the rows are equal:

$$\mathbf{p}(n) \to (0.12195, 0.18293, 0.25610, 0.12195, 0.31707).$$

In the next subsection we will show an easier way of finding the steady state pmf.

The random surfer model forms the basis for the PageRank algorithm that was introduced by Google to rank the importance of a page in the Web. The rank of a page is given by the steady state probability of the page in the Markov chain model. The size of the state space in this Markov chain is in the billions of pages!

### 11.2.3 Steady State Probabilities

Example 11.11 is typical of Markov chains that settle into stationary behavior after the process has been running for a long time. As $n \to \infty$, the $n$-step transition probability matrix approaches a matrix in which all the rows are equal to the same pmf, that is,

$$p_{ij}(n) \to \pi_j \quad \text{for all } i. \tag{11.17a}$$

We can express the above in matrix notation as:

$$P^n \to \mathbf{1}\boldsymbol{\pi} \tag{11.17b}$$

where $\mathbf{1}$ is a column vector of all 1's, that is, $\mathbf{1}^{\mathrm{T}} = (1, 1, \dots)$ and $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)$. From Eq. (11.16a), the convergence of $P^n$ implies the convergence of the state pmf's:

$$p_j(n) = \sum_i p_{ij}(n) p_i(0) \to \sum_i \pi_j p_i(0) = \pi_j. \tag{11.18}$$

We say that the system reaches "equilibrium" or "steady state."

We can find the pmf $\boldsymbol{\pi} \triangleq \{\pi_j\}$ in Eq. (11.18) (when it exists) by noting that as $n \to \infty$, $p_j(n) \to \pi_j$ and $p_i(n-1) \to \pi_i$, so Eq. (11.15) approaches

$$\pi_j = \sum_i p_{ij} \pi_i, \tag{11.19a}$$

which in matrix notation is

$$\boldsymbol{\pi} = \boldsymbol{\pi} P. \tag{11.19b}$$

Equation (11.19b) is underdetermined and requires the normalization equation:

$$\sum_i \pi_i = 1. \tag{11.19c}$$

We refer to $\boldsymbol{\pi}$ as the **stationary state pmf** of the Markov chain. If we start the Markov chain with initial state pmf $\mathbf{p}(0) = \boldsymbol{\pi}$, then by Eqs. (11.16b) and (11.19b) we have that the state probability vector

$$\mathbf{p}(n) = \boldsymbol{\pi} P^n = \pi \qquad \text{for all } n.$$

The resulting process is a stationary random process as defined in Section 9.6, since the probability of the sequence of states $i_0, i_1, \ldots, i_n$ starting at time $k$ is, by Eq. (11.5),

$$\begin{aligned} P[X_{n+k} = i_n, \ldots, X_k = i_0] &= P[X_{n+k} = i_n \,|\, X_{n+k-1} = i_{n-1}] \ldots P[X_{1+k} = i_1 \,|\, X_k = i_0] P[X_k = i_0] \\ &= P[X_{n+k} = i_n \,|\, X_{n+k-1} = i_{n-1}] \ldots P[X_{1+k} = i_1 \,|\, X_k = i_0] \pi_{i_0} \\ &= p_{i_{n-1},i_n} \cdots p_{i_0,i_1} \pi_{i_0}, \end{aligned}$$

which is independent of the initial time $k$. Thus the probabilities are independent of the choice of time origin, and the process is stationary.

---

**Example 11.13**

Find the stationary state pmf in Example 11.6.

Equation (11.19a) gives

$$\begin{aligned} \pi_0 &= (1 - \alpha)\pi_0 + \beta \pi_1 \\ \pi_1 &= \alpha \pi_0 + (1 - \beta)\pi_1, \end{aligned}$$

which imply that $\alpha \pi_0 = \beta \pi_1 = \beta(1 - \pi_0)$ since $\pi_0 + \pi_1 = 1$. Thus

$$\pi_0 = \frac{\beta}{\alpha + \beta} = \frac{2}{3} \qquad \pi_1 = \frac{\alpha}{\alpha + \beta} = \frac{1}{3}.$$

---

In this section we have shown the typical behavior of many Markov chains where the $n$-step transition probabilities and the state probabilities converge to constants that are independent of the initial conditions. These constant probabilities are found by solving the set of linear equations (11.19). It is worth noting, however, that not all Markov chains settle into stationary behavior where the process "forgets" the initial conditions. For example, the binomial counting process (Example 9.15) with $p > 0$ grows steadily so that for any fixed $j$, $p_j(n) \to 0$ as $n \to \infty$. The following example shows two atypical situations where the initial conditions determine the behavior for all time.

---

**Example 11.14 Two-State Process with Atypical Behavior**

Consider the two-state process with state transition diagram shown in Fig. 11.2(a). In Example 11.10 we found that the two-state process settles into steady state behavior so long as $|1 - \alpha - \beta| < 1$. Let's see what happens when this condition is not satisfied.

Consider first the case where $\alpha = \beta = 1$, and suppose that we start the process in state 0, that is, $p_0(0) = 1$. The state probabilities at time $n$ are:

$$\mathbf{p}(n) = (p_0(0), 1 - p_0(0)P^n = (1, 0)\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^n.$$

The process in this case alternates between state 0 at even time instants and state 1 at odd time instants. $P^n$ does not converge, and instead alternates assuming the values $P$ and $P^2 = I$. The state probability vector alternates between the values $(1, 0)$ and $(0, 1)$ so it does not exhibit convergence.

Now consider the case $\alpha = \beta = 0$, and suppose again that we start the process in state 0, that is, $p_0(0) = 1$. The state probabilities at time $n$ are:

$$\mathbf{p}(n) = (1, 0)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^n = (1, 0) \text{ for all } n.$$

In this case, the process remains fixed at state 0, which was selected at the initial time instant. Note that the process would have remained fixed at state 1 if state 1 had been selected initially. The state probability vector remains fixed at $(1, 0)$ if the initial state was 0 or $(0, 1)$ if the initial state was 1. In this case, both $P^n$ and $\mathbf{p}(n)$ converge immediately but to values that are determined by the initial condition.

---

The previous example demonstrates that we need to identify the conditions under which the state probability of Markov chains will converge to a stationary pmf that is found from Eq. (11.19). This is the topic of the next section.

## 11.3 CLASSES OF STATES, RECURRENCE PROPERTIES, AND LIMITING PROBABILITIES

In this section we take a closer look at the relation between the behavior of a Markov chain and its transition probability matrix. First we see that the states of a discrete-time Markov chain can be divided into one or more separate classes and that these classes can be of several types. We then show that the long-term behavior of a Markov chain is related to the types of its state classes. Figure 11.5 summarizes the types of classes to which a state can belong and identifies the associated long-term behavior.

### 11.3.1 Classes of States

We say that **state $j$ is accessible from state $i$** if for some $n \geq 0$, $p_{ij}(n) > 0$, that is, if there is a sequence of transitions from $i$ to $j$ that has nonzero probability. We say that **states $i$ and $j$ communicate** if they are accessible to each other; we then write $i \leftrightarrow j$. Note that a state communicates with itself since $p_{ii}(0) = 1$.

If state $i$ communicates with state $j$ and state $j$ communicates with state $k$, that is, $i \leftrightarrow j$ and $j \leftrightarrow k$, then state $i$ communicates with $k$. To see this, note that $i \leftrightarrow j$ implies that there is a nonzero probability path from $i$ to $j$ and $j \leftrightarrow k$ implies that there is a subsequent nonzero probability path from $j$ to $k$. The combined paths form a nonzero probability path from $i$ to $k$. A nonzero probability path in the reverse direction exists for the same reasons.

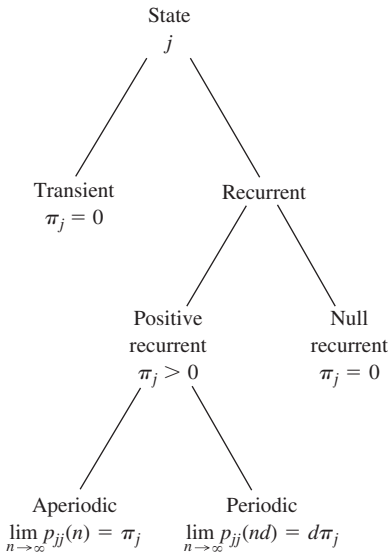



**FIGURE 11.5**
Classification of states and associated long-term behavior. The proportion of time spent in state $j$ is denoted by $\pi_j$.

We say that two states belong to the same **class** if they communicate with each other. Note that two different classes of states must be disjoint since having a state in common would imply that the states from both classes communicate with each other. Thus *the states of a Markov chain consist of one or more disjoint communication classes.* A Markov chain that consists of a single class is said to be **irreducible**.

---

**Example 11.15**

Figure 11.6(a) shows the state transition diagram for a Markov chain with three classes: $\{0\}$, $\{1, 2\}$, and $\{3\}$.

---

**Example 11.16**

Figure 11.6(b) shows the state transition diagram for a Markov chain with one class: $\{0, 1, 2, 3\}$. Thus the chain is irreducible.

---

**Example 11.17 Binomial Counting Process**

Figure 11.6(c) shows the state transition diagram for a binomial counting process. It can be seen that the classes are: $\{0\}$, $\{1\}$, $\{2\}, \ldots$.

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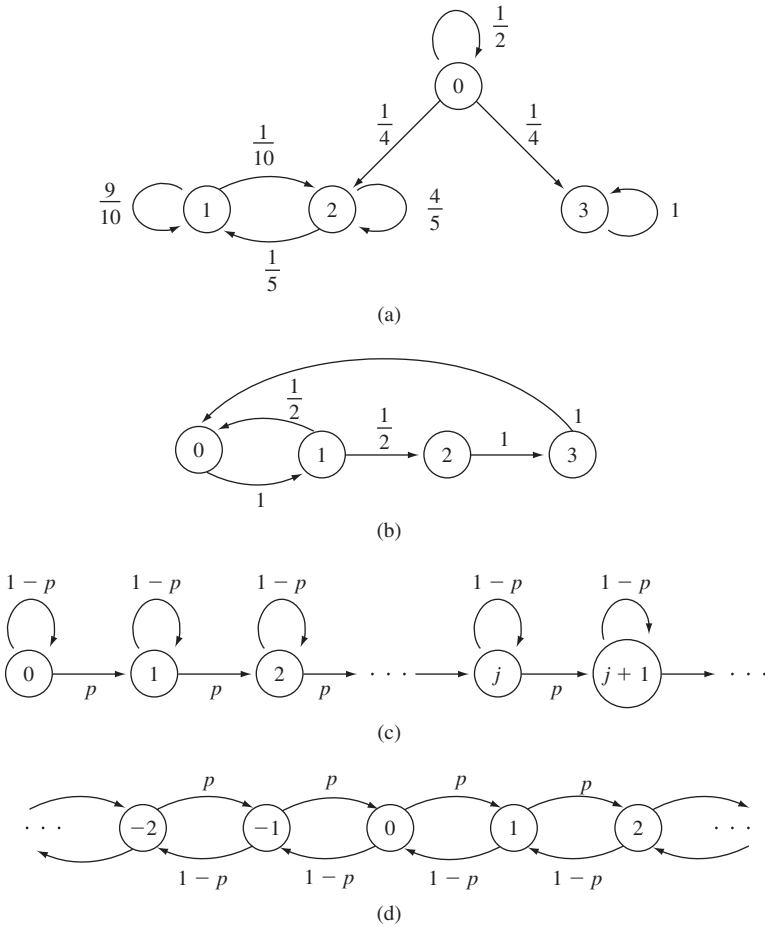



**FIGURE 11.6**
(a) A three-class Markov chain. (b) A periodic Markov chain. (c) A binomial counting process. (d) The random walk process.

**Example 11.18 Random Walk**

Figure 11.6(d) shows the state transition diagram for the random walk process. If $p > 0$, then the process has only one class, $\{0, \pm 1, \pm 2, \dots\}$, so it is irreducible.

### 11.3.2 Recurrence Properties

Suppose we start a Markov chain in state $i$. State $i$ is said to be **recurrent** if the process returns to the state with probability one, that is,

$$f_i = P[\text{ever returning to state } i] = 1. \tag{11.20a}$$

State $i$ is said to be **transient** if

$$f_i < 1. \tag{11.20b}$$

If we start the Markov chain in a recurrent state $i$, then the state reoccurs an infinite number of times. If we start the Markov chain in a transient state, the state does not reoccur after some finite number of returns. Each reoccurrence of the state can be viewed as a failure in a Bernoulli trial. The probability of failure is $f_i$. Thus the number of returns to state $i$ terminating with a success (no return) is a geometric random variable with mean $(1 - f_i)^{-1}$. If $f_i < 1$, then the probability of an infinite number of successes is zero. Therefore a transient state reoccurs only a finite number of times.

Let $X_n$ denote the Markov chain with initial state $i$, $X_0 = i$. Let $I_i(X)$ be the indicator function for state $i$, that is, $I_i(X)$ is equal to 1 if $X = i$ and equal to 0 otherwise. The expected number of returns to state $i$ is then

$$E\left[\sum_{n=1}^{\infty} I_i(X_n) \,|\, X_0 = i\right] = \sum_{n=1}^{\infty} E[I_i(X_n) \,|\, X_0 = i] = \sum_{n=1}^{\infty} p_{ii}(n) \tag{11.21}$$

since by Example 4.16

$$E[I_i(X_n) \,|\, X_0 = i] = P[X_n = i \,|\, X_0 = i] = p_{ii}(n).$$

A state is recurrent if and only if it reoccurs an infinite number of times, thus from Eq. (11.21) *state i is recurrent if and only if*

$$\sum_{n=1}^{\infty} p_{ii}(n) = \infty. \tag{11.22}$$

Similarly, *state i is transient if and only if*

$$\sum_{n=1}^{\infty} p_{ii}(n) < \infty. \tag{11.23}$$

---

**Example 11.19**

In Example 11.15 (Fig. 11.6a), state 0 is transient since $p_{00}(n) = (1/2)^n$, so

$$\sum_{n=1}^{\infty} p_{00}(n) = \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots = 1 < \infty.$$

On the other hand, if the process were started in state 1, we would have the two-state process discussed in Example 11.10. For such a process we found that

$$p_{11}(n) = \frac{\beta + \alpha(1 - \alpha - \beta)^n}{\alpha + \beta} = \frac{1/2 + 1/4(7/10)^n}{3/4}$$

so that

$$\sum_{n=1}^{\infty} p_{11}(n) = \sum_{n=1}^{\infty}\left(\frac{2}{3} + \frac{(7/10)^n}{3}\right) = \infty.$$

Therefore state 1 is recurrent.

---

**Example 11.20 Binomial Counting Process**

In the binomial counting process all the states are transient since $p_{ii}(n) = (1 - p)^n$ so that for $p > 0$,

$$\sum_{n=1}^{\infty} p_{ii}(n) = \sum_{n=1}^{\infty} (1 - p)^n = \frac{1 - p}{p} < \infty.$$

**Example 11.21 Random Walk**

Consider state zero in the random walk process in Fig. 11.6(d). The state reoccurs in $2n$ steps if and only if $n$ +1s and $n$ −1s occur during the $2n$ steps. This occurs with probability

$$p_{00}(2n) = \binom{2n}{n} p^n (1 - p)^n.$$

Stirling's formula for $n!$ can be used to show that

$$\binom{2n}{n} p^n (1 - p)^n \sim \frac{(4p(1 - p))^n}{\sqrt{\pi n}},$$

where $a_n \sim b_n$ when $\lim_{n\to\infty} a_n/b_n = 1$.

Thus Eq. (11.21) for state 0 is

$$\sum_{n=1}^{\infty} p_{00}(2n) \sim \sum_{n=1}^{\infty} \frac{(4p(1 - p))^n}{\sqrt{\pi n}}.$$

If $p = 1/2$, then $4p(1 - p) = 1$ and the series diverges. It then follows that state 0 is recurrent. If $p \neq 1/2$, then $(4p(1 - p)) < 1$, and the above series converges. This implies that state 0 is transient. Thus when $p = 1/2$, the random walk process maintains a precarious balance about 0. As soon as $p \neq 1/2$, a positive or negative drift is introduced and the process grows towards $\pm\infty$.

*Recurrence and transience are class properties*: If a state $i$ is recurrent, then all states in its class are recurrent; if a state is transient, then all the states in its class are transient. If state $i$ is recurrent, then all states in its class will be visited eventually as the process forever returns to state $i$ over and over again. Indeed all other states in its class will appear an infinite number of times.

To show the recurrence class property, let $i$ be a recurrent state and let $j$ be another state in the class, then $i \leftrightarrow j$, and there are probabilities $p_{ji}(m) > 0$ and $p_{ij}(l) > 0$ that corresponds to nonzero probability paths that lead from $j$ to $i$ in $m$ steps, and back from $i$ to $j$ in $l$ steps. We can identify many nonzero probability paths that go from $j$ to $j$ by splicing the above two paths to recurrent paths for state $i$: go from $j$ to $i$ using the above path; then from $i$ to $i$ using an $n$-step recurrent path; then back from $i$ to $j$ using the above path. The probabilities for these paths provide a lower bound to the recurrence probabilities for $j$:

$$\sum_k p_{jj}(k) > \sum_n p_{ji}(m) p_{ii}(n) p_{ij}(l) = p_{ji}(m) p_{ij}(l) \sum_n p_{ii}(n) = \infty,$$

since state $i$ is recurrent. This implies that state $j$ is also recurrent. Now suppose that state $i$ is transient, and let $j$ be another state in its class. State $j$ cannot be recurrent, for this would imply that $i$ is recurrent, in contradiction to our assumption. Therefore $j$ must be transient.

*If a Markov chain is irreducible then either all its states are transient or all its states are recurrent.* If the Markov chain has a finite state space, it is impossible for all of its states to be transient. At least some of the states must occur an infinite number of times as time progresses, implying that all states are recurrent. Therefore, *the states of a finite-state, irreducible Markov chain are all recurrent*. If the state space is countably infinite, then all the states can be transient. The random walk with $p \neq 1/2$ provides an example of such a Markov chain.

The structure of the state transition diagram and the associated nonzero transition probabilities can impose periodicity in the realizations of a discrete-time Markov chain. We say that state $i$ has **period** $d$ if it can only reoccur at times that are multiples of $d$, that is, $p_{ii}(n) = 0$ whenever $n$ is not a multiple of $d$, where $d$ is the largest integer with this property. We say that state $i$ is **aperiodic** if it has period $d = 1$.

*Periodicity is a class property*, that is, all states in a class have the same period. An irreducible Markov chain is said to be **aperiodic** if the states in its single class have period one. An irreducible Markov chain is said to be **periodic** if its states have period $d > 1$.

To show that periodicity is a class property, suppose that state $i$ has period $d$ and let $j$ be another state in the same class. Since $i \leftrightarrow j$, there are probabilities $p_{ji}(m) > 0$ and $p_{ij}(l) > 0$ that corresponds to paths that lead from $j$ to $i$ in $m$ steps, and back from $i$ to $j$ in $l$ steps. We can create a path from $j$ to $j$ by splicing the $m$-step path for $j$ to $i$ with the $l$-step path from $i$ to $j$; this path has length $m + l$ and probability $p_{ji}(m)p_{ij}(l) > 0$. The length $m + l$ must be divisible by $d'$, the period of state $j$. Now create multiple paths from $j$ to $j$ by attaching the above two paths to nonzero probability paths that go from $i$ to $i$ in $n$ steps. These paths have length $m + l + n$ and probability $p_{ji}(m)p_{ii}(n)p_{ij}(l) > 0$. All these paths go from $j$ to $j$ so $m + n + l$ must be divisible by $d'$. We already showed that $m + l$ is divisible by $d'$, so we have that $n$ must also be divisible by $d'$. But $n$ can be the length of any path that goes from $i$ to $i$, and so $d$, the period of state $i$, is the largest value that divides all such $n$. This implies that $d'$ must divide $d$. By reversing the roles of state $i$ and state $j$, the same series of arguments imply that $d$ must divide $d'$. Thus $d = d'$ and state $i$ and state $j$ have the same period.

---

**Example 11.22 Two-State Process with Atypical Behavior**

Characterize the two "atypical" Markov chains in Example 11.14.

In the case where $\alpha = \beta = 1$, Fig. 11.2(a) shows that we have a single communication class with period $d = 2$. This explains why the process alternates between state 0 at even time instants and state 1 at odd time instants

In the case $\alpha = \beta = 0$, we have two communication classes: $\{0\}$ and $\{1\}$. The selection of the initial state at $t = 0$ effectively picks one of the two classes, and the process remains in that class forever.

---

**Example 11.23**

In Example 11.15 (Fig 11.6a), all the states have the property that $p_{ii}(n) > 0$ for $n = 1, 2, \ldots$. Therefore all three classes in the Markov chain have period 1.

**Example 11.24**

In the Markov chain in Fig 11.6(b), the states 0 and 1 can reoccur at time $2, 4, 6, \ldots$ and states 2 and 3 at times $4, 6, 8, \ldots$. Therefore the Markov chain has period 2.

**Example 11.25**

In the random walk process in Fig 11.6(d), a state reoccurs when the number of successes $(+1\text{s})$ equals the number of failures $(-1\text{s})$. This can only happen after an even number of steps. The process therefore has period 2.

Figure 11.7(a) summarizes the possible structures that can be encountered for Markov chains. In the case of irreducible finite-state Markov chains, all states in the single class must be recurrent and the class can either be aperiodic or periodic. If a finite-state

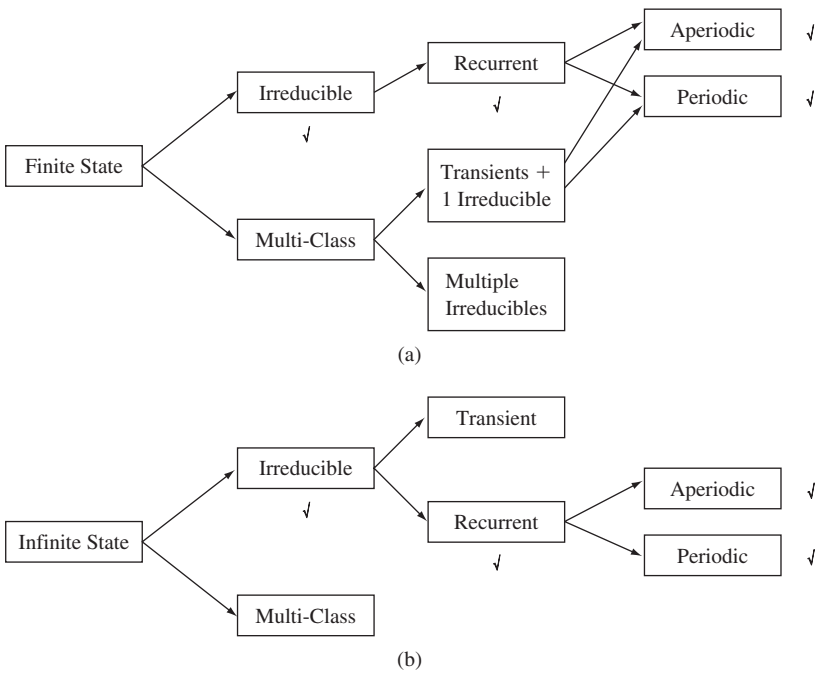



**FIGURE 11.7**
Possible structures for Markov chains.

Markov chain consists of multiple transient classes and a single irreducible class, then the chain will eventually settle in the states of the irreducible class. Thus in the long-run the behavior is the same as that of an irreducible chain. A finite-state Markov chain with multiple irreducible classes will eventually enter and remain thereafter in one of the irreducible classes. Over the long run, the chain will exhibit the behavior of an irreducible Markov chain with the given class of states. Thus the case of multi-irreducible classes can be viewed as a two stage random experiment in which the first stage involves selecting one of the irreducible classes.

Figure 11.7(b) summarizes the possible structures for Markov chains with infinite state space. The major difference from the finite case is that an irreducible class can have all of its states be transient. Consequently when a chain has multiple classes it is now possible for the chain to enter and remain in a class that is either transient or recurrent.

### 11.3.3 Limiting Probabilities

If all the states in a Markov chain are transient, then all the state probabilities approach zero as $n \to \infty$. If a Markov chain has some transient classes and some recurrent classes, as in Fig. 11.6(a), then eventually the process enters and remains thereafter in one of the recurrent classes. Therefore we can concentrate on individual recurrent classes when studying the limiting probabilities of a chain. For this reason we assume in this section that we are dealing with an irreducible Markov chain.

Suppose we start a Markov chain in a *recurrent* state $i$ at time $n = 0$. Let $T_i(1), T_i(1) + T_i(2), \ldots$ be the times when the process returns to state $i$, where $T_i(k)$ is the time that elapses between the $(k-1)$th and $k$th returns (see Fig. 11.8). The $T_i$ form an iid sequence since each return time is independent of previous return times.

The proportion of time spent in state $i$ after $k$ returns to $i$ is

$$\text{proportion of time in state } i = \frac{k}{T_i(1) + T_i(2) + \cdots + T_i(k)}. \tag{11.24}$$

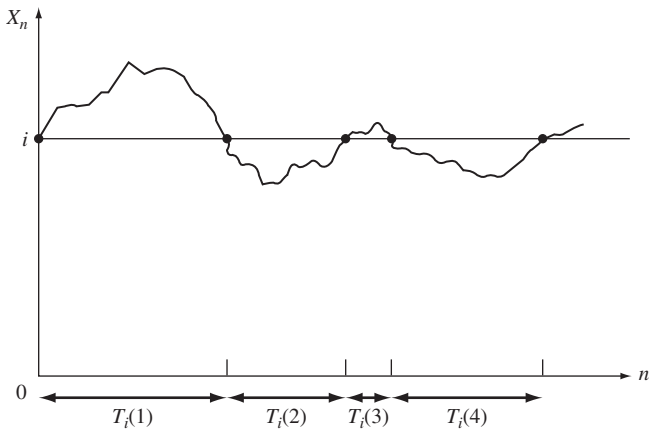



**FIGURE 11.8**
Recurrence times for state *i*.

Since the state is recurrent, the process returns to state $i$ an infinite number of times. Thus the law of large numbers implies that, with probability one, the reciprocal of the above expression approaches the **mean recurrence time** $E[T_i]$ so the long-term proportion of time spent in state $i$ approaches

$$\text{proportion of time in state } i \to \frac{1}{E[T_i]} = \pi_i, \tag{11.25}$$

where $\pi_i$ *is the long-term proportion of time spent in state i.*

If $E[T_i] < \infty$, then we say that state $i$ is **positive recurrent**. Equation (11.25) then implies that

$$\pi_i > 0 \qquad \text{if state } i \text{ is positive recurrent.}$$

If $E[T_i] = \infty$, then we say that state $i$ is **null recurrent**. Equation (11.25) then implies that

$$\pi_i = 0 \qquad \text{if state } i \text{ is null recurrent.}$$

It can be shown that positive and null recurrence are class properties.

Positive recurrent, aperiodic states are called **ergodic**. Once a Markov chain enters an ergodic state, then the process will remain in the state's class forever. Furthermore the process will visit all states in the class sufficiently frequently that the long-term proportion of time in a given state will be governed by Eq. (11.25) and approach a nonzero value. Thus the process will reveal its underlying state probabilities through time averages. Given our previous discussion on ergodicity in Chapter 9, it is not surprising that an **ergodic Markov chain** is defined as an irreducible, aperiodic, positive recurrent Markov chain.

---

**Example 11.26**

The process in Fig. 11.6(b) returns to state 0 in two steps with probability 1/2 and in four steps with probability 1/2. Therefore the mean recurrence time for state 0 is

$$E[T_0] = \frac{1}{2}(2) + \frac{1}{2}(4) = 3.$$

Therefore state 0 is positive recurrent and the long-term proportion of time spent in state 0 is

$$\pi_0 = \frac{1}{3}.$$

---

**Example 11.27 Random Walk**

In Example 11.21 it was shown that the random walk process is recurrent if $p = 1/2$. However, the mean recurrence time can be shown to be infinite when $p = 1/2$ (Feller, 1968, p. 314). Thus all the states in the chain are null recurrent.

---

The $\pi_j$'s in Eq. (11.25) satisfy the equations that define the stationary state pmf:

$$\pi_j = \sum_i \pi_i P_{ij} \qquad \text{for all } j \tag{11.26a}$$

and

$$1 = \sum_i \pi_i. \tag{11.26b}$$

To see this, note that since $\pi_i$ is the proportion of time spent in state $i$, then $\pi_i P_{ij}$ is the proportion of time in which state $j$ follows $i$. If we sum over all $i$, we then obtain the long-term proportion of time in state $j$, $\pi_j$.

---

**Example 11.28**

The stationary state pmf for the periodic Markov chain in Fig. 11.6(b) is found from Eqs. (11.26a) and (11.26b):

$$\begin{aligned}
\pi_0 &= \frac{1}{2}\pi_1 + \pi_3 \\
\pi_1 &= \pi_0 \\
\pi_2 &= \frac{1}{2}\pi_1 \\
\pi_3 &= \pi_2.
\end{aligned}$$

These equations imply that $\pi_1 = \pi_0$ and $\pi_2 = \pi_3 = \pi_0/2$. Since the probabilities must add to one, we obtain

$$\pi_1 = \pi_0 = \frac{1}{3} \qquad \text{and} \qquad \pi_2 = \pi_3 = \frac{1}{6}.$$

Note that $\pi_0 = 1/3$ was obtained for the mean recurrence time in Example 11.26.

---

In Section 11.2 we found that for certain Markov chains, the $n$-step transition matrix approaches a fixed matrix of equal rows as $n \to \infty$ (see Eq. 11.17). We also saw that the rows of this limiting matrix consisted of a pmf that satisfied Eqs. (11.26a) and (11.26b). We are now ready to state under what conditions this occurs.

---

**Theorem 1**[2]

For an irreducible aperiodic Markov chain exactly one of the following assertions holds:

**(i)** All states are transient or all states are null recurrent; $p_{ij}(n) \to 0$ as $n \to \infty$ for all $i$ and $j$ and there exists no stationary pmf

**(ii)** All states are positive recurrent, so

$$\lim_{n\to\infty} p_{ij}(n) = \pi_j \quad \text{for all } j \tag{11.27}$$

where $\{\pi_j, j = 1, 2, 3, \dots\}$ is the unique stationary pmf solution to Eq. (11.26ab).

---

[2]A proof to Theorem 1 is given by [Ross, pp. 108–110].

Theorem 1 states that for ergodic Markov chains, the $n$-step transition probabilities approach constant values given by the steady state pmf. Note that Eq. (11.27) can be written in matrix form as shown in Eq. (11.17b). From Eq. (11.18), it then follows that the state probabilities approach steady state values that are independent of the initial conditions. These steady state probabilities correspond to the stationary probabilities obtained by solving Eq. (11.26ab), and thus correspond to the long-term proportion of time spent in a given state. Theorem 1 also states that if the irreducible Markov chain is transient or null recurrent, then a stationary pmf solution to Eq. (11.26ab) does not exist. This implies that when we do find a solution, and the chain is irreducible and aperiodic, then the Markov chain must be positive recurrent and hence ergodic.

### Example 11.29 Age of a Device

Consider a Markov Chain that counts the age of a device in service at the end of each day. At the end of each day, the device either increases its age by 1 (with probability $a$) or fails and returns to the "1" state (with probability $1 - a$). A failed device is replaced at the beginning of the next day and the age counting processes is resumed. Determine whether the Markov chain has a stationary distribution.

The state transition diagram for the Markov chain is shown in Fig. 11.9. If $a > 0$, then every state $i$ can access any state $i + 1$, and consequently any state $i$ can access any state $j > i$. In addition every state $i$ can access state 1. This implies that there is a nonzero probability path between any two states, and so the Markov chain is irreducible. State 1 can reoccur in intervals of $1, 2, 3, 4, \ldots$, and so state 1 has period 1. Therefore all the states have period 1 and the Markov chain is aperiodic.

The equations for the stationary probabilities are:

$$\pi_1 = (1 - a)\pi_1 + (1 - a)\pi_2 + \cdots = (1 - a)(\pi_1 + \pi_2 + \cdots) = 1 - a$$

$$\pi_{i+1} = a\pi_i \quad \text{for} \quad i \geq 1.$$

By a simple induction argument we can show that:

$$\pi_i = (1 - a)a^{i-1} \quad \text{for} \quad i \geq 1.$$

Therefore the Markov chain is positive recurrent and has this stationary pmf.

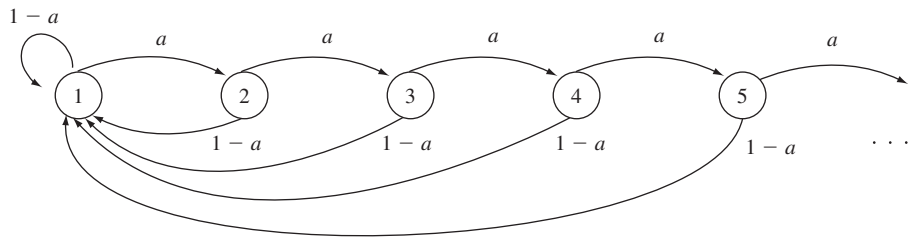



**FIGURE 11.9**
Age of a device.

### Example 11.30 Google PageRank Algorithm

In Example 11.12 we showed the basic approach for ranking Web pages according to an associated Markov chain. The approach included a strategy to deal with the case where users become trapped in a page with no outgoing links, i.e., page 2 in Fig. 11.4(a). The approach, however, is not sufficient to ensure that the Markov chain is irreducible and aperiodic. For example, in Fig. 11.4(b) users can also become trapped in the periodic class $\{4, 5\}$. This poses a problem for the rank algorithm which uses the power of the transition probability matrix to obtain the stationary pmf. To deal with this problem, the PageRank algorithm also assumes that each time a new page is selected, the procedure in Example 11.12 is used with probability $\alpha$, but otherwise (with probability $1 - \alpha$) any of all possible Web pages is selected with equal probability. The value $\alpha = 0.85$ is usually cited as appropriate. The modified ranking method then has a transition probability matrix that is aperiodic and irreducible and the conditions of Theorem 1 are satisfied.

For the example in Fig. 11.4(b) we have:

$$P = (0.85)\begin{bmatrix} 0 & 1/2 & 1/2 & 0 & 0 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/3 & 1/3 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1/2 & 0 & 1/2 \end{bmatrix} + (0.15)\begin{bmatrix} 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0300 & 0.4550 & 0.4550 & 0.0300 & 0.0300 \\ 0.2000 & 0.2000 & 0.2000 & 0.2000 & 0.2000 \\ 0.3133 & 0.3133 & 0.0300 & 0.3133 & 0.0300 \\ 0.0300 & 0.0300 & 0.0300 & 0.0300 & 0.8800 \\ 0.0300 & 0.0300 & 0.4550 & 0.0300 & 0.4550 \end{bmatrix}.$$

The matrix $P$ has a stationary state pmf given by:

$$\mathbf{p}(n) = (0.13175, 0.18772, 0.24642, 0.13173, 0.30239).$$

See [Langville] for more details on the PageRank algorithm.

For periodic processes, we have the following result.

### Theorem 2

For an irreducible, periodic, and positive recurrent Markov chain with period $d$,

$$\lim_{n\to\infty} p_{jj}(nd) = d\pi_j \text{ for all } j \tag{11.28}$$

where $\pi_j$ is the unique nonnegative solution of Eqs. (11.26a) and (11.26b).

As before, $\pi_j$ represents the proportion of time spent in state $j$. However, the fact that state $j$ is constrained to occur at multiples of $d$ steps implies that the probability of occurrence of the state $j$ is $d$ times greater at the allowable times and zero elsewhere.

**Example 11.31**

In Examples 11.26 and 11.28 we found that the long-term proportion of time spent in state 0 is $\pi_0 = 1/3$. If we start in state 0, then only even states can occur at even time instants. Thus at these even time instants the probability of state 0 is 2/3 and of state 2 is 1/3. At odd time instants, the probabilities of states 0 and 2 are zero.

Theorems 1 and 2 only address the most important cases of irreducible, periodic and aperiodic Markov chains indicated by the checkmarks in Fig. 11.7. The following example considers a case not covered by Theorems 1 and 2.

**Example 11.32 Markov Chain with Multiple Irreducible Classes**

Does the Markov chain in Fig. 11.6(a) have a unique stationary pmf?

The equations for the stationary probabilities are:

$$\begin{aligned} p_0 &= 1/2p_0 \\ p_1 &= 9/10p_1 + 1/5p_2 \\ p_2 &= 1/4p_0 + 1/10p_1 + 4/5p_2 \\ p_3 &= 1/4p_0 + p_3. \end{aligned}$$

The first equation implies that $p_0 = 0$, which reduces the fourth equation to $p_3 = p_3$, which imposes no constraints on $p_3$. The middle two equations are equivalent and both imply that $p_1 = 2p_2$. The normalization condition requires that $1 = p_1 + p_2 + p_3 = 3p_2 + p_3$. Therefore the equations are underdetermined and there are many solutions with the form: $(0, 2p_2, p_2, 1 - 3p_2)$ where $0 \le p_2 \le 1/3$.

Now let's approach the problem according to its three classes: $\{0\}$, $\{1, 2\}$, and $\{3\}$. The first class is transient and the other two classes are recurrent. Suppose the initial state is 3, then the process remains in that state forever. The stationary pmf for class $\{3\}$ by itself is $(0, 0, 0, 1)$. If the initial state is 1 or 2, then the process remains in this class forever; the stationary pmf for this class in isolation is $(0, 2/3, 1/3, 0)$. Finally if the initial state is 0, then the process will eventually leave and enter one of the other two classes with equal probability. In the general case, if the initial state is selected according to the pmf $(p_0(0), p_1(0), p_2(0), p_3(0))$ then the class $\{1, 2\}$ will be entered with probability $1/2\, p_0(0) + p_1(0) + p_2(0)$, and class $\{3\}$ will be entered with probability $1/2\, p_0(0) + p_3(0)$. The stationary pmf would then have the form:

$$\begin{aligned} &\{1/2\, p_0(0) + p_1(0) + p_2(0)\}(0, 2/3, 1/3, 0) + \{1/2\, p_0(0) + p_3(0)\}\,(0, 0, 0, 1) \\ &\quad = \gamma(0, 2/3, 1/3, 0) + (1 - \gamma)(0, 0, 0, 1) \\ &\quad = (0, 2\gamma/3, \gamma/3, 1 - \gamma). \end{aligned}$$

If we let $\gamma/3 = p_2$ we see that this solution has the form we derived before.

For example, suppose the initial pmf was $(0, 1/3, 1/6, 1/2)$, then this pmf satisfies the condition for a stationary pmf and the repeated multiplication by $P$ will yield the same pmf. In this sense this multiclass Markov chain has a stationary pmf. Note however that the relative frequencies of the states depend on which irreducible class is actually entered. Thus if we record long-term average frequencies we will observe either $(0, 2/3, 1/3, 0)$ or $(0, 0, 0, 1)$. The stationary pmf

does not correspond to either of these two pmf's; instead the stationary pmf gives us the expected value of the two pmf's:

$$(0, 1/3, 1/6, 1/2) = 1/2(0, 2/3, 1/3, 0) + 1/2(0, 0, 0, 1)$$

where 1/2 is the probability of entering the two irreducible classes for this choice of initial pmf.

Example 11.32 illustrates the behavior of multiclass finite-state Markov chain. In these chains the process will eventually enter and remain forever in one of its recurrent classes. Each recurrent class can be considered as a separate irreducible Markov chain with its own stationary pmf. The multiclass Markov chain will then have stationary pmf's that depend on the stationary pmf's of its constituent recurrent classes according to the initial state probabilities. These multiclass Markov chains are not ergodic since the relative frequencies of the states do not correspond to the stationary pmf.

If a multiclass chain has infinite state space, then the situation discussed above can occur as a special case: the process initially works its way through transient classes and eventually settles in one of a number of ergodic classes. However, in general, it is possible for some or all of the classes to be transient and/or null recurrent. In such case the process may never settle into stationary behavior.

## 11.4 CONTINUOUS-TIME MARKOV CHAINS

In Section 11.2 we saw that the transition probability matrix determines the behavior of a discrete-time Markov chain. In this section we see that the same is true for continuous-time Markov chains.

The joint pmf for $k + 1$ arbitrary time instants of a Markov chain is given by Eq. (11.3):

$$\begin{aligned} P[X(t_{k+1}) = x_{k+1}, X(t_k) = x_k, \ldots, X(t_1) = x_1] \\ = P[X(t_{k+1}) = x_{k+1} \,|\, X(t_k) = x_k] \cdots \\ \times P[X(t_2) = x_2 \,|\, X(t_1) = x_1] P[X(t_1) = x_1]. \end{aligned} \quad (11.29)$$

This result holds regardless of whether the process is discrete-time or continuous-time. In the continuous-time case, Eq. (11.29) requires that we know the transition probabilities from an arbitrary time $s$ to an arbitrary time $s + t$:

$$P[X(s + t) = j \,|\, X(s) = i] \qquad t \geq 0.$$

We assume here that the transition probabilities depend only on the difference between the two times:

$$P[X(s + t) = j \,|\, X(s) = i] = P[X(t) = j \,|\, X(0) = i] = p_{ij}(t) \qquad t \geq 0, \text{ all } s. \quad (11.30)$$

We say that $X(t)$ has **homogeneous transition probabilities**.

Let $P(t) = \{p_{ij}(t)\}$ *denote the matrix of transition probabilities in an interval of length t.* Since $p_{ii}(0) = 1$ and $p_{ij}(0) = 0$ for $i \neq j$, we have

$$P(0) = I, \tag{11.31}$$

where $I$ is the identity matrix.

---

**Example 11.33 Poisson Process**

For the Poisson process, the transition probabilities satisfy

$$\begin{aligned} p_{ij}(t) &= P[j - i \text{ events in } t \text{ seconds}] \\ &= p_{0,\,j-i}(t) \\ &= \frac{(\alpha t)^{j-i}}{(j-i)!} e^{-\alpha t} \qquad j \geq i. \end{aligned}$$

Therefore

$$P(t) = \begin{bmatrix} e^{-\alpha t} & \alpha t e^{-\alpha t} & (\alpha t)^2 e^{-\alpha t}/2! & \cdot & \cdots \\ 0 & e^{-\alpha t} & \alpha t e^{-\alpha t} & (\alpha t)^2 e^{-\alpha t}/2! & \cdots \\ 0 & 0 & e^{-\alpha t} & \alpha t e^{-\alpha t} & \cdots \\ \cdot & \cdot & \cdot & \cdot & \cdots \end{bmatrix}.$$

As $t$ approaches zero, $e^{-\alpha t} \approx 1 - \alpha t$. Thus for a small time interval $\delta$,

$$P(\delta) \approx \begin{bmatrix} 1 - \alpha\delta & \alpha\delta & 0 & \cdots \\ 0 & 1 - \alpha\delta & \alpha\delta & \cdots \\ 0 & 0 & 1 - \alpha\delta & \cdots \\ \cdot & \cdot & \cdot & \cdots \end{bmatrix},$$

where all terms of order $\delta^2$ or higher have been neglected. Thus the probability of more than one transition in a very short time interval is negligible. Note that this is consistent with the assumptions made in deriving the Poisson process in Section 9.4.

---

**Example 11.34 Random Telegraph**

In the random telegraph example, the process $X(t)$ changes with each occurrence of an event in a Poisson process. From Eqs. (9.40) and (9.41) we see that the transition probabilities are as follows:

$$\begin{aligned} P[X(t) = a \,|\, X(0) = a] &= \frac{1}{2}\{1 + e^{-2\alpha t}\} \\ P[X(t) = a \,|\, X(0) = b] &= \frac{1}{2}\{1 - e^{-2\alpha t}\} \qquad \text{if } a \neq b. \end{aligned}$$

Thus the transition probability matrix is

$$P(t) = \begin{bmatrix} 1/2\{1 + e^{-2\alpha t}\} & 1/2\{1 - e^{-2\alpha t}\} \\ 1/2\{1 - e^{-2\alpha t}\} & 1/2\{1 + e^{-2\alpha t}\} \end{bmatrix}.$$

---

### 11.4.1 State Occupancy Times

Since the random telegraph signal changes polarity with each occurrence of an event in a Poisson process, it follows that the time spent in each state is an exponential random variable. It turns out that this is a property of the **state occupancy time** for all continuous-time Markov chains, that is: $X(t)$ *remains at a given value* (*state*) *for an exponentially distributed random time.* To see why, let $T_i$ be the time spent in a state $i$. The probability of spending more than $t$ seconds in this state is then

$$P[T_i > t].$$

Now suppose that the process has already been in state $i$ for $s$ seconds; then the probability of spending $t$ more seconds in this state is

$$P[T_i > t + s \,|\, T_i > s] = P[T_i > t + s \,|\, X(s') = i, 0 \leq s' \leq s],$$

since the $\{T_i > s\}$ implies that the system has been in state $i$ during the time interval $(0, s)$. The Markov property implies that if $X(s) = i$, then the past is irrelevant and we can view the system as being restarted in state $i$ at time $s$:

$$P[T_i > t + s \,|\, T_i > s] = P[T_i > t]. \tag{11.32}$$

Only the exponential random variable satisfies this memoryless property (see Section 4.4). Thus the time spent in state $i$ is an exponential random variable with some mean $1/v_i$:

$$P[T_i > t] = e^{-v_i t}. \tag{11.33}$$

The *mean state occupancy time* $1/v_i$ will usually be different for each state.

The above result provides us with another way of looking at continuous-time Markov chains. Each time a state, say $i$, is entered, an exponentially distributed state occupancy time $T_i$ is selected. When the time is up, the next state $j$ is selected according to a *discrete-time* Markov chain, with transition probabilities $\tilde{q}_{ij}$. Then the new state occupancy time is selected according to $T_j$, and so on.[3] We call $\tilde{q}_{ij}$ an **embedded Markov chain**. We will see in the last part of this section that the properties of the continuous-time Markov chain depends on the class properties of its embedded chain.

---

**Example 11.35**

The random telegraph signal in Example 11.34 spends an exponentially distributed time with mean $1/\alpha$ in each state. When a transition occurs, the transition is always from the present state to the only other state, thus the embedded Markov chain is

$$\tilde{q}_{00} = 0 \qquad \tilde{q}_{01} = 1$$
$$\tilde{q}_{10} = 1 \qquad \tilde{q}_{11} = 0.$$

---

[3]This view of Markov chains is useful in setting up computer simulation models of Markov chain processes.

### 11.4.2 Transition Rates and Time-Dependent State Probabilities

Consider the transition probabilities in a very short time interval of duration $\delta$ seconds. The probability that the process remains in state $i$ during the interval is

$$\begin{aligned} P[T_i > \delta] &= e^{-v_i\delta} \\ &= 1 - \frac{v_i\delta}{1!} + \frac{v_i^2\delta^2}{2!} - \cdots \\ &= 1 - v_i\delta + o(\delta), \end{aligned}$$

where $o(\delta)$ denotes terms that become negligible relative to $\delta$ as $\delta$ approaches zero.[4] The exponential distributions of the state occupancy times imply that the probability of two or more transitions in an interval of duration $\delta$ is $o(\delta)$. Thus for small $\delta$, $p_{ii}(\delta)$ is approximately equal to the probability that the process remains in state $i$ for $\delta$ seconds:

$$\begin{aligned} p_{ii}(\delta) &= P[T_i > \delta] + o(\delta) \\ &= 1 - v_i\delta + o(\delta) \end{aligned}$$

or equivalently,

$$1 - p_{ii}(\delta) = v_i\delta + o(\delta). \tag{11.34}$$

We call $v_i$ the *rate at which the process $X(t)$ leaves state $i$.*

Once the process leaves state $i$, it will enter state $j$ with probability $\widetilde{q}_{ij}$, where $\widetilde{q}_{ij}$ is the transition probability of the embedded Markov chain. Thus

$$\begin{aligned} p_{ij}(\delta) &= (1 - p_{ii}(\delta))\widetilde{q}_{ij} \\ &= v_i\widetilde{q}_{ij}\delta + o(\delta) \\ &= \gamma_{ij}\delta + o(\delta). \end{aligned} \tag{11.35a}$$

We call $\gamma_{ij} = v_i\widetilde{q}_{ij}$ the *rate at which the process $X(t)$ enters state $j$ from state $i$.* For completeness, we define $\gamma_{ii} = -v_i$, so that by Eq. (11.34),

$$p_{ii}(\delta) - 1 = \gamma_{ii}\delta + o(\delta). \tag{11.35b}$$

If we divide both sides of Eqs. (11.35a) and (11.35b) by $\delta$ and take the limit $\delta \to 0$, we obtain

$$\lim_{\delta\to 0}\frac{p_{ij}(\delta)}{\delta} = \gamma_{ij} \qquad i \neq j \tag{11.36a}$$

and

$$\lim_{\delta\to 0}\frac{p_{ii}(\delta) - 1}{\delta} = \gamma_{ii}, \tag{11.36b}$$

since

$$\lim_{\delta\to 0}\frac{o(\delta)}{\delta} = 0,$$

because $o(\delta)$ is of order higher than $\delta$.

[4] A function $g(h)$ is said to be $o(h)$ if $\lim_{h\to 0} g(h)/h = 0$, that is, $g(h)$ goes to zero faster than $h$ does.

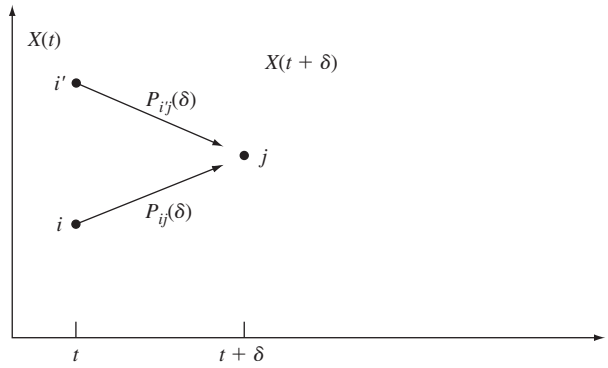



**FIGURE 11.10**
Transitions into state *j*.

We are now ready to develop a set of equations for finding the state probabilities at time $t$, which will be denoted by

$$p_j(t) \triangleq P[X(t) = j].$$

For $\delta > 0$, we have (see Fig. 11.10)

$$\begin{aligned} p_j(t + \delta) &= P[X(t + \delta) = j] \\ &= \sum_i P[X(t + \delta) = j \,|\, X(t) = i] P[X(t) = i] \\ &= \sum_i p_{ij}(\delta) p_i(t). \end{aligned} \tag{11.37}$$

If we subtract $p_j(t)$ from both sides, we obtain

$$p_j(t + \delta) - p_j(t) = \sum_{i \neq j} p_{ij}(\delta) p_i(t) + (p_{jj}(\delta) - 1) p_j(t). \tag{11.38}$$

If we divide by $\delta$, apply Eqs. (11.36a) and (11.36b) and let $\delta \to 0$, we obtain

$$p_j'(t) = \sum_i \gamma_{ij} p_i(t). \tag{11.39}$$

Equation (11.39) is a form of the **Chapman–Kolmogorov equations** for continuous-time Markov chains. To find $p_j(t)$ we need to solve this system of differential equations with initial conditions specified by the initial state pmf $\{p_j(0), j = 0, 1, \dots\}$.

Note that if we solve Eq. (11.39) under the assumption that the state at time zero was $i$, that is, with initial condition $p_i(0) = 1$ and $p_j(0) = 0$ for all $j \neq i$, then the solution is actually $p_{ij}(t)$, the $ij$ component of $P(t)$. Thus Eq. (11.39) can also be used to find the transition probability matrix.

### Example 11.36 A Simple Queueing System

A queueing system alternates between two states. In state 0, the system is idle and waiting for a customer to arrive. This idle time is an exponential random variable with mean $1/\alpha$. In state 1, the system is busy servicing a customer. The time in the busy state is an exponential random variable with mean $1/\beta$. Find the state probabilities $p_0(t)$ and $p_1(t)$ in terms of the initial state probabilities $p_0(0)$ and $p_1(0)$.

The system moves from state 0 to state 1 at a rate $\alpha$, and from state 1 to state 0 at a rate $\beta$:

$$\gamma_{00} = -\alpha \qquad \gamma_{01} = \alpha$$
$$\gamma_{10} = \beta \qquad \gamma_{11} = -\beta.$$

Equation (11.39) then gives

$$p_0'(t) = -\alpha p_0(t) + \beta p_1(t)$$
$$p_1'(t) = \alpha p_0(t) - \beta p_1(t).$$

Since $p_0(t) + p_1(t) = 1$, the first equation becomes

$$p_0'(t) = -\alpha p_0(t) + \beta(1 - p_0(t)),$$

which is a first-order differential equation:

$$p_0'(t) + (\alpha + \beta)p_0(t) = \beta \qquad p_0(0) = p_0.$$

The general solution of this equation is

$$p_0(t) = \frac{\beta}{\alpha + \beta} + Ce^{-(\alpha+\beta)t}.$$

We obtain $C$ by setting $t = 0$ and solving in terms of $p_0(0)$; then we find

$$p_0(t) = \frac{\beta}{\alpha + \beta} + \left(p_0(0) - \frac{\beta}{\alpha + \beta}\right)e^{-(\alpha+\beta)t}$$

and

$$p_1(t) = \frac{\alpha}{\alpha + \beta} + \left(p_1(0) - \frac{\alpha}{\alpha + \beta}\right)e^{-(\alpha+\beta)t}.$$

Note that as $t \to \infty$,

$$p_0(t) \to \frac{\beta}{\alpha + \beta} \qquad \text{and} \qquad p_1(t) \to \frac{\alpha}{\alpha + \beta}.$$

Thus as $t \to \infty$, the state probabilities approach constant values that are independent of the initial state probabilities.

### Example 11.37 The Poisson Process

Find the state probabilities for the Poisson process.

The Poisson process moves only from state $i$ to state $i + 1$ at a rate $\alpha$.

Thus

$$\gamma_{ii} = -\alpha \qquad \text{and} \qquad \gamma_{i,\,i+1} = \alpha.$$

Equation (11.39) then gives

$$p_0'(t) = -\alpha p_0(t) \qquad \text{for } j = 0$$
$$p_j'(t) = -\alpha p_j(t) + \alpha p_{j-1}(t) \qquad \text{for } j \geq 1.$$

The initial condition for the Poisson process is $p_0(0) = 1$, so the solution for the $j = 0$ equation is

$$p_0(t) = e^{-\alpha t}.$$

The equation for $j = 1$ is

$$p_1'(t) = -\alpha p_1(t) + \alpha e^{-\alpha t} \qquad p_1(0) = 0,$$

which is also a first-order differential equation for which the solution is

$$p_1(t) = \frac{\alpha t}{1!} e^{-\alpha t}.$$

It can be shown by an induction argument that the solution of the state $j$ equation is

$$p_j(t) = \frac{(\alpha t)^j}{j!} e^{-\alpha t}.$$

For any fixed time $t$, the sum of $\{p_j(t)\}$ is one. Note however, that for any $j$, $p_j(t) \to 0$ as $t \to \infty$. Figure 11.11 shows how the pmf drifts to higher values as time progresses. Thus for the Poisson process, the probability of any finite state approaches zero as $t \to \infty$. This is consistent with the fact that the process grows steadily with time.

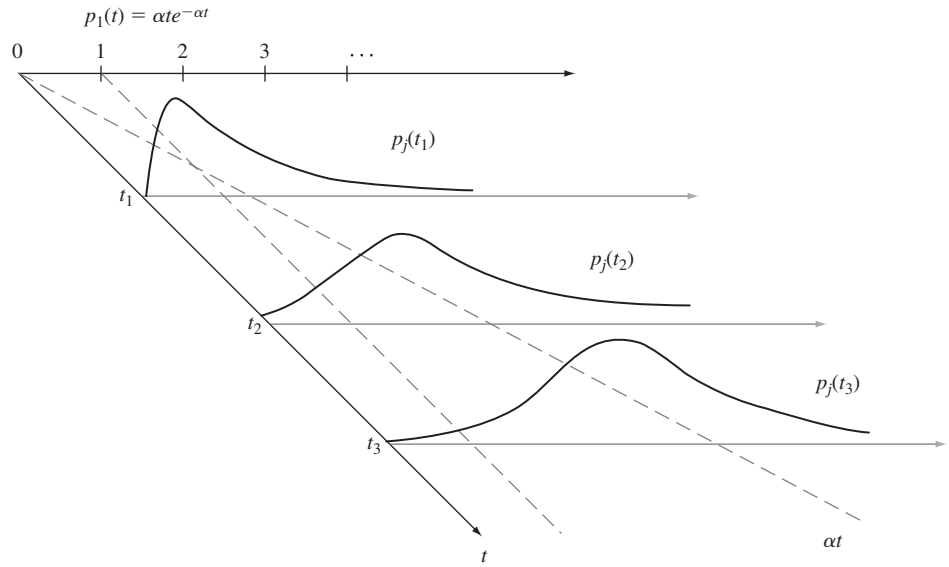



**FIGURE 11.11**
State pmf of Poisson process vs. time.

### 11.4.3 Steady State Probabilities and Global Balance Equations

As $t \to \infty$, the state probabilities in the two-state queueing system in Example 11.36 converge to a pmf that does not depend on the initial conditions. This is typical of systems that reach "equilibrium" or "steady state." For such a system, $p_j(t) \to p_j$ and $p_j'(t) \to 0$, so Eq. (11.39) becomes

$$0 = \sum_i \gamma_{ij} p_i \qquad \text{for all } j, \tag{11.40a}$$

or equivalently, recalling that $\gamma_{jj} = -v_j$,

$$v_j p_j = \sum_{i \neq j} \gamma_{ij} p_i \qquad \text{for all } j, \tag{11.40b}$$

where

$$\sum_j p_j = 1. \tag{11.40c}$$

Equation (11.40b) can be rewritten as follows:

$$p_j \left( \sum_{i \neq j} \gamma_{ji} \right) = \sum_{i \neq j} \gamma_{ij} p_i \tag{11.40d}$$

since

$$v_j = \sum_{i \neq j} \gamma_{ji}.$$

The system of linear equations given by Eq. (11.40b) or (11.40d) are called the **global balance equations**. These equations state that at equilibrium, the rate of probability flow out of state $j$, namely $v_j p_j$, is equal to the rate of flow into state $j$, as shown in Fig. 11.12. By solving this set of linear equations we can obtain the stationary state pmf of the system (when it exists).[5]

We refer to $\mathbf{p} = \{p_i\}$ as the **stationary state pmf** of the Markov chain. Since $\mathbf{p}$ satisfies Eq. (11.39), if we start the Markov chain with initial state pmf given by $\mathbf{p}$, then the state probabilities will be

$$p_i(t) = p_i \qquad \text{for all } t.$$

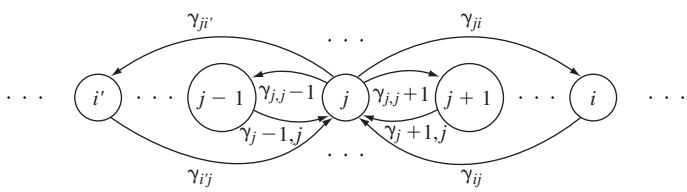



**FIGURE 11.12**
Global balance of probability flows.

[5]The last part of this section discusses conditions under which the stationary pmf exists.

The resulting process is a stationary random process as defined in Section 9.6 since the probability of the sequence of states $i_0, i_1, \ldots, i_n$ at times $t < t_1 + t < \cdots < t_n + t$ is, by Eq. (11.29),

$$\begin{aligned} P[X(t) = i_0, X(t_1 + t) = i_1, \ldots, X(t_n + t) = i_n] \\ &= P[X(t_n + t) = i_n \,|\, X(t_{n-1} + t) = i_{n-1}] \cdots \\ &\quad \times P[X(t_1 + t) = i_1 \,|\, X(t) = i_0] P[X(t) = i_0]. \end{aligned}$$

The transition probabilities depend only on the difference between the associated times. Thus the above joint probability depends on the choice of origin only through $P[X(t) = i_0]$. But $P[X(t) = i_0] = p_{i_0}$ for all $t$. Therefore we conclude that the above joint probability is independent of the choice of time origin and thus that the process is stationary.

---

**Example 11.38**

Find the stationary state pmf for the two-state queueing system discussed in Example 11.36.

Equation (11.40b) for this system gives

$$\alpha p_0 = \beta p_1 \qquad \text{and} \qquad \beta p_1 = \alpha p_0.$$

Noting that $p_0 + p_1 = 1$, we obtain

$$p_0 = \frac{\beta}{\alpha + \beta} \qquad \text{and} \qquad p_1 = \frac{\alpha}{\alpha + \beta}.$$

---

**Example 11.39 The M/M/1 Single-Server Queueing System**

Consider a queueing system in which customers are served one at a time in order of arrival. The time between customer arrivals is exponentially distributed with rate $\lambda$, and the time required to service a customer is exponentially distributed with rate $\mu$. Find the steady state pmf for the number of customers in the system.

The state transition rates are as follows. Customers arrive at a rate $\lambda$, so

$$\gamma_{i,i+1} = \lambda \qquad i = 0, 1, 2, \ldots.$$

When the system is nonempty, customers depart at the rate $\mu$. Thus

$$\gamma_{i,i-1} = \mu \qquad i = 1, 2, 3, \ldots.$$

The transition rate diagram is shown in Fig. 11.13. The global balance equations are

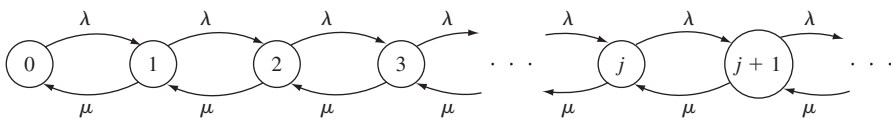



**FIGURE 11.13**
Transition rate diagram for M/M/1 queueing system.

$$\lambda p_0 = \mu p_1 \qquad \text{for } j = 0 \tag{11.41a}$$

$$(\lambda + \mu)p_j = \lambda p_{j-1} + \mu p_{j+1} \qquad \text{for } j = 1, 2, \ldots. \tag{11.41b}$$

We can rewrite Eq. (11.41b) as follows:

$$\lambda p_j - \mu p_{j+1} = \lambda p_{j-1} - \mu p_j \qquad \text{for } j = 1, 2, \ldots,$$

which implies that

$$\lambda p_{j-1} - \mu p_j = \text{constant} \qquad \text{for } j = 1, 2, \ldots. \tag{11.42}$$

Equation (11.42) with $j = 1$ and Eq. (11.41a) together imply that

$$\text{constant} = \lambda p_0 - \mu p_1 = 0.$$

Thus Eq. (11.42) becomes

$$\lambda p_{j-1} = \mu p_j,$$

or equivalently,

$$p_j = \rho p_{j-1} \qquad j = 1, 2, \ldots$$

and by a simple induction argument

$$p_j = \rho^j p_0,$$

where $\rho = \lambda/\mu$. We obtain $p_0$ by noting that the sum of the probabilities must be one:

$$1 = \sum_{j=0}^{\infty} p_j = (1 + \rho + \rho^2 + \cdots)p_0 = \frac{1}{1 - \rho} p_0,$$

where the series converges if and only if $\rho < 1$.

Thus

$$p_j = (1 - \rho)\rho^j \qquad j = 0, 1, 2, \ldots. \tag{11.43}$$

This queueing system is discussed in detail in Section 12.3.

The condition for the existence of a steady state solution has a simple explanation. The condition $\rho < 1$ is equivalent to

$$\lambda < \mu,$$

that is, the rate at which customers arrive must be less than the rate at which the system can process them. Otherwise the queue builds up without limit as time progresses.

---

### Example 11.40 A Birth-and-Death Process

A **birth-and-death process** is a Markov chain in which only transitions between adjacent states occur as shown in Fig. 11.14. The single-server queueing system discussed in Example 11.39 is an example of a birth-and-death process.

The global balance equations for a general birth-and-death process are

$$\lambda_0 p_0 = \mu_1 p_1 \qquad j = 0 \tag{11.44a}$$

$$\lambda_j p_j - \mu_{j+1} p_{j+1} = \lambda_{j-1} p_{j-1} - \mu_j p_j \qquad j = 1, 2, \ldots. \tag{11.44b}$$

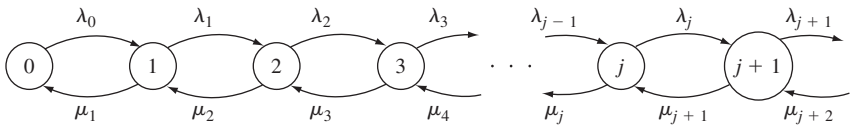



**FIGURE 11.14**
Transition rate diagram for general birth-and-death process.

As in the previous example, it then follows that

$$p_j = r_j p_{j-1} \qquad j = 1, 2, \ldots$$

and

$$p_j = r_j r_{j-1} \ldots r_1 p_0 \qquad j = 1, 2, \ldots, \tag{11.45}$$

where $r_j = (\lambda_{j-1})/\mu_j$. If we define

$$R_j = r_j r_{j-1} \ldots r_1 \qquad \text{and} \qquad R_0 = 1,$$

then $p_0$ is found from

$$1 = \left( \sum_{j=0}^{\infty} R_j \right) p_0.$$

If the series in the above equation converges, then the stationary pmf is given by

$$p_j = \frac{R_j}{\sum_{i=0}^{\infty} R_i}. \tag{11.46}$$

If the series does not converge, then a stationary pmf does not exist, and $p_j = 0$ for all $j$. In Chapter 12, we will see that many useful queueing systems can be modeled by birth-and-death processes.

### 11.4.4 Limiting Probabilities for Continuous-Time Markov Chains

We saw above that a continuous-time Markov chain $X(t)$ can be viewed as consisting of a sequence of states determined by some discrete-time Markov chain $X_n$ with transition probabilities $\widetilde{q}_{ij}$ and a corresponding sequence of exponentially distributed state occupancy times. In this section we use this approach to investigate the limiting probabilities of continuous-time Markov chains.

First we consider the construction of stationary solutions for $X(t)$ from the steady state solutions of $X_n$. Suppose that the embedded Markov chain $X_n$ is irreducible and positive recurrent, so that Eq. (11.25) holds. Let $N_i(n)$ denote the number of times state $i$ occurs in the first $n$ transitions, and let $T_i(j)$ denote the occupancy time the $j$th time state $i$ occurs. The proportion of time spent by $X(t)$ in state $i$ after the first $n$ transitions is

$$\frac{\text{time spent in state } i}{\text{time spent in all states}} = \frac{\sum_{j=1}^{N_i(n)} T_i(j)}{\sum_i \sum_{j=1}^{N_i(n)} T_i(j)}$$

$$= \frac{\frac{N_i(n)}{n} \frac{1}{N_i(n)} \sum_{j=1}^{N_i(n)} T_i(j)}{\sum_i \frac{N_i(n)}{n} \frac{1}{N_i(n)} \sum_{j=1}^{N_i(n)} T_i(j)}. \tag{11.47}$$

As $n \to \infty$, by Eqs. (11.25) and (11.26ab), with probability one,

$$\frac{N_i(n)}{n} \to \pi_i, \tag{11.48}$$

the stationary pmf of the embedded Markov chain. In addition, we also have that $N_i(n) \to \infty$ as $n \to \infty$, so that by the strong law of large numbers, with probability one,

$$\frac{1}{N_i(n)} \sum_{j=1}^{N_i(n)} T_i(j) \to E[T_i] = 1/v_i, \tag{11.49}$$

where we have used the fact that the state occupancy time in state $i$ has mean $1/v_i$. Similarly the denominator in Eq. (11.47) must approach $\left(\sum \pi_j/v_j\right)$. Equations (11.48) and (11.49) when applied to Eq. (11.47) imply that if $\sum \pi_j/v_j < \infty$, with probability one, the long-term proportion of time spent in state $i$ approaches

$$p_i = \frac{\pi_i/v_i}{\sum_j \pi_j/v_j} = c\pi_i/v_i, \tag{11.50}$$

where $\pi_j$ is the unique pmf solution to

$$\pi_j = \sum_i \pi_i \widetilde{q}_{ij} \qquad \text{for all } j \tag{11.51}$$

and $c$ is a normalization constant.

We obtain the global balance equation, Eq. (11.40b), by substituting $\pi_i = v_i p_i/c$ from Eq. (11.50) and $\widetilde{q}_{ij} = \gamma_{ij}/v_i$ into Eq. (11.51):

$$v_j p_j = \sum_{i \neq j} p_i \gamma_{ij} \qquad \text{for all } j.$$

Thus the $p_i$'s are the unique solution of the global balance equations.

We have proved the following result:

---

**Theorem 3**

Assume a time-continuous Markov chain, for which the embedded Markov chain is irreducible and positive recurrent with stationary pmf $\{\pi_j\}$ and $\sum_j \pi_j/v_j < \infty$, then the following assertions hold:

**(i)** $\lim_{t\to\infty} p_j(t) = p_j$ for all $j$;

**(ii)** The solution $\{p_i\}$ is unique and satisfies Eqs. (11.40bc);

**(iii)** For each $j$, $p_j$ is the long-term proportion of time spent in state $j$.

---

Now assume that we know that the Markov chain is irreducible and that we have a solution $\{p_j\}$ to the global balance equations (11.40bc):

$$p_j v_j = \sum_{i\neq j} p_i \gamma_{ij}.$$

Substituting Eq. (11.50) into the above equation

$$c\pi_j = \left(\frac{c\pi_j}{v_j}\right) v_j = \sum_{i\neq j}\left(\frac{c\pi_i}{v_i}\right)\gamma_{ij} = c\sum_{i\neq j}\pi_i\left(\frac{\gamma_{ij}}{v_i}\right) = c\sum_{i\neq j}\pi_i \widetilde{q}_{ij}$$

implies that the following choice of $\{\pi_j\}$ gives a solution for the stationary pmf of the embedded Markov chain:

$$\pi_j = \frac{p_j v_j}{\sum_i p_i v_i}.$$

Note that we must require that the denominator be finite. From Theorem 1 in Section 11.4, if there is a stationary pmf then it is unique and positive recurrent. Furthermore the construction of $\{\pi_j\}$ from the $\{p_j\}$ ensures that $p_j$ is the long-term proportion of time in state $j$ as well as the limiting state probability for $X(t)$.

We have shown the following theorem:

---

**Theorem 4**

Assume a time-continuous Markov chain, for which the embedded Markov chain is irreducible. Suppose that $\{p_j\}$ is a solution to the global balance equations (11.40bc), and that $\sum_j \pi_j v_j < \infty$, then the following assertions hold:

**(i)** The solution $\{p_i\}$ is unique;

**(ii)** $\lim_{t\to\infty} p_j(t) = p_j$ for all $j$;

**(iii)** For each $j$, $p_j$ is the long-term proportion of time spent in state $j$;

**(iv)** The embedded Markov chain is positive recurrent.

---

**Example 11.41**

In the two-state system in Example 11.36,

$$[\widetilde{q}_{ij}] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The equation $\boldsymbol{\pi} = \boldsymbol{\pi}[\widetilde{q}_{ij}]$ implies that

$$\pi_0 = \pi_1 = \frac{1}{2}.$$

In addition, $v_0 = \alpha$ and $v_1 = \beta$. Thus

$$p_0 = \frac{1/2(1/\alpha)}{1/2(1/\alpha + 1/\beta)} = \frac{\beta}{\alpha + \beta}$$

and

$$p_1 = \frac{\alpha}{\alpha + \beta}.$$

---

## *11.5 TIME-REVERSED MARKOV CHAINS

We now consider the random process that results when we play a Markov chain backwards in time. We will see that the resulting process is also a Markov chain and so develop another method for obtaining the stationary probabilities of the forward and reverse processes. The insights gained by looking at the reverse process prove useful in developing certain results in queueing theory in Chapter 12.

Let $X_n$ be a stationary ergodic Markov chain[6] with one-step transition probability matrix $P = \{p_{ij}\}$ and stationary state pmf $\{\pi_j\}$. Consider the dependence of $X_{n-1}$, the "future" in the reverse process, on $X_n, X_{n+1}, \ldots, X_{n+k}$, the "present and past":

$$\begin{aligned} P[X_{n-1} = j \,|\, X_n = i, X_{n+1} = i_1, \ldots, X_{n+k} = i_k] &= \frac{P[X_{n-1} = j, X_n = i, X_{n+1} = i_1, \ldots, X_{n+k} = i_k]}{P[X_n = i, X_{n+1} = i_1, \ldots, X_{n+k} = i_k]} \\ &= \frac{\pi_j p_{ji} p_{i,i_1} \cdots p_{i_{k-1},i_k}}{\pi_i p_{i,i_1} \cdots p_{i_{k-1},i_k}} \\ &= \frac{\pi_j p_{ji}}{\pi_i} \\ &= P[X_{n-1} = j \,|\, X_n = i]. \end{aligned} \tag{11.52}$$

[6]That is, let it be an irreducible, aperiodic, stationary Markov chain.

The above equations show that *the **time-reversed process** is also a Markov chain with one-step transition probabilities*

$$P[X_{n-1} = j \mid X_n = i] = q_{ij} = \frac{\pi_j p_{ji}}{\pi_i}. \tag{11.53}$$

Since $X_n$ is irreducible and aperiodic, its stationary state probabilities $\pi_j$ represent the proportion of time that the state is in state $j$. This proportion of time does not depend on whether one goes forward or backward in time, so $\pi_j$ must also be the stationary pmf for the reverse process. Thus *the forward and reverse process must have the same stationary pmf.*

---

**Example 11.42**

Suppose that a new light bulb is put in use at day $n = 0$, and suppose that each time a light bulb fails it is replaced the next day. Let $X_n$ be the age of the light bulb (in days) at the end of day $n$. If $a_i$ is the probability that the lifetime $L$ of a light bulb is $i$ days, then the probability that the light bulb fails on day $j$ given that it has not failed up to then is

$$b_j = \frac{P[L = j]}{P[L \geq j]} = \frac{a_j}{\sum_{k=j}^{\infty} a_k} \qquad j = 1, 2, \ldots.$$

Thus the transition probabilities for $X_n$ are

$$\begin{aligned} p_{i,i+1} &= 1 - b_i \qquad i = 1, 2, \ldots \\ p_{i1} &= b_i \qquad\quad i = 1, 2, \ldots \\ p_{ij} &= 0 \qquad\quad \text{otherwise.} \end{aligned}$$

Figure 11.15(a) shows the state transition diagram of $X_n$, and Fig. 11.16(a) shows a typical sample function that consists of a sawtooth-shaped function that increases linearly and then falls abruptly to one when a light bulb fails.

Figure 11.16(b) shows a sample function of the reverse process from which we deduce that the state transition diagram must be as shown in Fig. 11.15(b). The transition probabilities for the reverse process are obtained from Eq. (11.53):

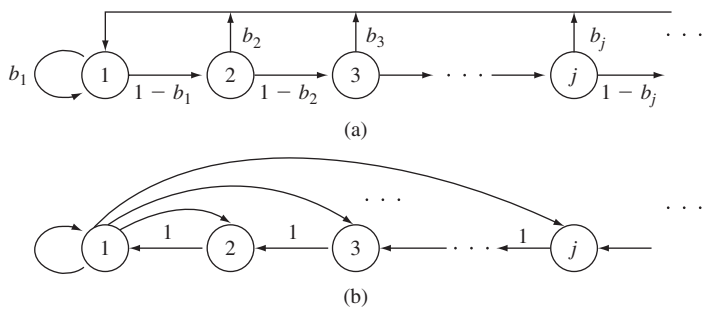



**FIGURE 11.15**
(a) Transition diagram for age of a renewal process. (b) Transition diagram for time-reversed process.

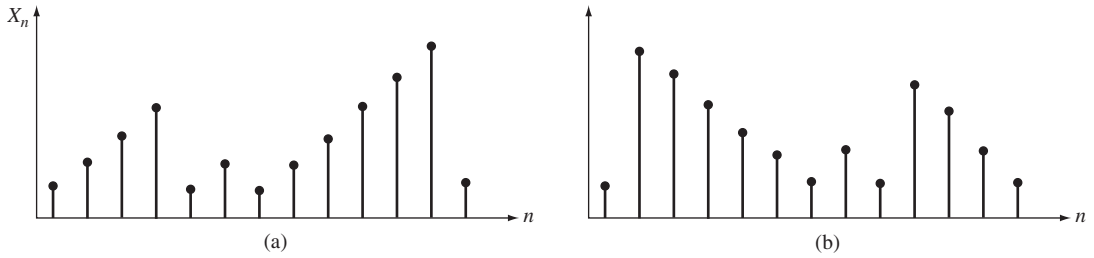



**FIGURE 11.16**
(a) Age of light bulb in use at time $n$. (b) Time-reversed process of $X_n$.

$$q_{i,i-1} = \frac{\pi_{i-1}}{\pi_i}(1 - b_{i-1}) \qquad i = 2, 3, 4, \ldots$$

$$q_{1,i} = \frac{\pi_i}{\pi_1} b_i \qquad i = 1, 2, \ldots$$

$$q_{i,j} = 0 \qquad \text{otherwise.}$$

For now we defer the problem of finding the stationary state probabilities $\pi_j$.

---

Example 11.42 shows that Eq. (11.53) provides us with conditions that must be satisfied by the stationary probabilities $\pi_j$. Suppose we were able to *guess* a pmf $\{\pi_j\}$ so that Eq. (11.53) holds, that is,

$$\pi_i q_{ij} = \pi_j p_{ji} \qquad \text{for all } i, j. \tag{11.54}$$

It then follows that $\{\pi_j\}$ is the stationary pmf. To see this, sum Eq. (11.54) over all $j$, then

$$\sum_j \pi_j p_{ji} = \pi_i \sum_j q_{ij} = \pi_i \qquad \text{for all } i. \tag{11.55}$$

But Eq. (11.55) is the condition for $\pi_j$ to be the stationary pmf for the forward process, thus $\pi_j$ is the stationary pmf. Equation (11.54) thus provides us with another method for finding the stationary pmf of a discrete-time Markov chain: *If we can guess a set of transition probabilities $q_{i,j}$ for the reverse process and a pmf $\pi_j$ so that Eq. (11.54) is satisfied, then it follows that the $\pi_j$ is the stationary pmf for the Markov chain and the $q_{i,j}$ are the transition probabilities for the reverse process.*

---

**Example 11.43**

The sample function of the reverse process in Example 11.42 suggests that for $i > 1$, the process moves from state $i$ to state $i - 1$ with probability one; that is,

$$q_{i,i-1} = \frac{\pi_{i-1}(1 - b_{i-1})}{\pi_i} = 1,$$

which implies that

$$\begin{aligned}\pi_i &= (1 - b_{i-1})\pi_{i-1} \qquad i = 2, 3, \ldots \\ &= (1 - b_{i-1})(1 - b_{i-2}) \cdots (1 - b_1)\pi_1. \end{aligned} \tag{11.56}$$

However, from Example 11.42 for $i \geq 2$,

$$(1 - b_{i-1}) = 1 - \frac{a_{i-1}}{\sum_{k=i-1}^{\infty} a_k} = \frac{\sum_{k=i}^{\infty} a_k}{\sum_{k=i-1}^{\infty} a_k},$$

so in Eq. (11.56), the denominator of $(1 - b_{i-1})$ cancels the numerator of $(1 - b_{i-2})$, the denominator of $(1 - b_{i-2})$ cancels the numerator of $(1 - b_{i-3})$, and so on. Thus

$$\pi_i = \left\{ \sum_{k=i}^{\infty} a_k \right\} \pi_1 = P[L \geq i]\pi_1 \qquad i = 2, 3, \ldots .$$

We obtain $\pi_1$ by using the fact that the probabilities sum to one:

$$1 = \pi_1 \sum_{i=1}^{\infty} P[L \geq i] = \pi_1 E[L],$$

where we have used Eq. (4.29) for $E[L]$. Thus

$$\pi_i = \frac{P[L \geq i]}{E[L]} \qquad i = 1, 2, \ldots . \tag{11.57}$$

### 11.5.1 Time-Reversible Markov Chains

A stationary ergodic Markov chain is said to be **reversible** if the one-step transition probability matrix of the forward and reverse processes are the same, that is, if

$$q_{ij} = p_{ij} \qquad \text{for all } i, j. \tag{11.58}$$

Equations (11.53) and (11.58) together imply that a Markov chain is reversible if and only if

$$\pi_i p_{ij} = \pi_j p_{ji} \qquad \text{for all } i, j. \tag{11.59}$$

Since $\pi_i$ and $\pi_j$ are the long-term proportion of transitions out of states $i$ and $j$, respectively, Eq. (11.59) implies that a chain is reversible if the proportion of transitions from $i$ to $j$ is equal to the proportion of transitions from $j$ to $i$.

#### Example 11.44 Discrete-Time Birth-and-Death Process

Figure 11.17 shows the state transition diagram for a discrete-time birth-and-death process with transition probabilities

$$\begin{aligned} p_{00} &= 0 \qquad & p_{01} &= 1 = a_0 & \\ p_{i,i+1} &= a_i \qquad & p_{i,i-1} &= 1 - a_i \qquad & i = 1, 2, \ldots \\ p_{ij} &= 0 \qquad & \text{otherwise.} & & \end{aligned}$$

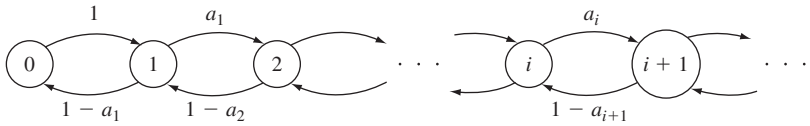



**FIGURE 11.17**
Transition diagram for a discrete-time birth-and-death process.

For any sample path, the number of transitions from $i$ to $i+1$ can differ by at most 1 from the number of transitions from $i+1$ to $i$ since the only way to return to $i$ is through $i+1$. Thus the long-term proportion of transitions from $i$ to $i+1$ is equal to that from $i+1$ to $i$. Since these are the only possible transitions, it follows that birth-and-death processes are reversible.

Equation (11.59) implies that

$$a_j\pi_j = (1 - a_{j+1})\pi_{j+1} \qquad j = 0, 1, 2, \ldots,$$

which allows us to write all the $\pi_j$'s in terms of $\pi_0$:

$$\pi_j = \left(\frac{a_{j-1}}{1 - a_j}\right)\cdots\left(\frac{a_0}{1 - a_1}\right)\pi_0 = \frac{a_{j-1}\cdots a_0}{(1 - a_j)\cdots(1 - a_1)}\pi_0 \triangleq R_j\pi_0. \tag{11.60}$$

The probability $\pi_0$ is found from $\quad 1 = \pi_0\sum_{j=0}^{\infty} R_j.$ (11.61)

The series in Eq. (11.61) must converge in order for $\pi_j$ to exist.

### 11.5.2 Time-Reversible Continuous-Time Markov Chains

Now consider a stationary, continuous-time Markov chain played backward in time. If $X(t) = i$ (i.e., the process is in state $i$ at time $t$), then the probability that the reverse process remains in state $i$ for an additional $s$ seconds is

$$\begin{aligned} P[X(t') = i, \quad t - s \le t' \le t \,|\, X(t) = i] &= \frac{P[X(t - s) = i, T_i > s]}{P[X(t) = i]} \\ &= \frac{P[X(t - s) = i]P[T_i > s]}{P[X(t) = i]} \\ &= P[T_i > s] = e^{-v_i s}, \end{aligned} \tag{11.62}$$

where $P[X(t - s) = i] = P[X(t) = i]$ because $X(t)$ is a stationary process, and where $T_i$ is the time spent in state $i$ for the forward process. Thus *the reverse process also spends an exponentially distributed amount of time with rate $v_i$ in state $i$.*

The jumps in the forward process $X(t)$ are determined by the embedded Markov chain $\widetilde{q}_{ij}$, so the jumps in the reverse process are determined by the discrete-time Markov chain corresponding to the time-reversed embedded Markov chain given by Eq. (11.53):

$$q_{ij} = \frac{\pi_j\widetilde{q}_{ji}}{\pi_i}. \tag{11.63}$$

It follows that the transition rates for the time-reversed continuous-time process are given by

$$\begin{aligned}\gamma'_{ij} &= v_i q_{ij} = \frac{\pi_j v_i \widetilde{q}_{ji}}{\pi_i} \\ &= \frac{v_i \pi_j \gamma_{ji}}{\pi_i v_j} = \frac{p_j \gamma_{ji}}{p_i},\end{aligned} \tag{11.64}$$

where we used the fact that $\widetilde{q}_{ji} = \gamma_{ji}/v_j$ and $p_j = c\pi_j/v_j$. In comparing Eq. (11.64) to Eq. (11.53), note that the transition rates $\gamma'_{ij}$ have simply replaced the transition probabilities $q_{ij}$ in going from the discrete-time to the continuous-time case.

The discussion that led to Eq. (11.54) provides us with another method for determining the stationary pmf $p_j$ of $X(t)$. *If we can guess a set of transition rates $\gamma'_{i,j}$ and a pmf $p_j$ such that*

$$p_i \gamma'_{i,j} = p_j \gamma_{j,i} \qquad \text{for all } i, j \tag{11.65a}$$

*and*

$$\sum_{j \neq i} \gamma_{i,j} = \sum_{j \neq i} \gamma'_{i,j} \qquad \text{for all } i, \tag{11.65b}$$

*then $p_j$ is the stationary pmf for $X(t)$ and $\gamma'_{i,j}$ are the transition rates for the reverse process.*

Since the state occupancy times in the forward and reverse processes are exponential random variables with the same mean, the continuous-time Markov chain $X(t)$ is reversible if and only if its embedded Markov chain is reversible. Equation (11.59) implies that the following condition must be satisfied:

$$\pi_i \widetilde{q}_{ij} = \pi_j \widetilde{q}_{ji} \qquad \text{for all } i, j, \tag{11.66}$$

where $\pi_j$ is the stationary pmf of the embedded Markov chain. Recall from Eq. (11.50) that $\pi_j = cv_j p_j$, where $p_j$ is the stationary pmf of $X(t)$. Substituting into Eq. (11.66), we obtain

$$p_i v_i \widetilde{q}_{ij} = p_j v_j \widetilde{q}_{ji},$$

which is equivalent to

$$p_i \gamma_{ij} = p_j \gamma_{ji}. \tag{11.67}$$

Thus we conclude that $X(t)$ *is reversible if and only if Eq. (11.67) is satisfied*. As in the discrete-time case, Eq. (11.67) can be interpreted as stating that the rate at which $X(t)$ goes from state $i$ to state $j$ is equal to the rate at which $X(t)$ goes from state $j$ to state $i$.

---

**Example 11.45 Continuous-Time Birth-and-Death Process**

Consider the general continuous-time birth-and-death process introduced in Example 11.40. The embedded Markov chain in this process is a discrete-time birth-and-death process of the type discussed in Example 11.44. It therefore follows that all continuous-time birth-and-death processes are time-reversible.

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In Chapter 12 we will see that the time reversibility of certain Markov chains implies some remarkable properties about the departure processes of queueing systems.

## 11.6 NUMERICAL TECHNIQUES FOR MARKOV CHAINS

In this section we present several numerical techniques that are useful in the analysis of Markov chains. The first part of the section presents methods for finding the stationary as well as transient solutions for the state probabilities of Markov chains. The second part of the section addresses the simulation of discrete-time and continuous-time Markov chains.

### 11.6.1 Stationary Probabilities of Markov Chains

The most basic calculation with *finite-state discrete-time Markov chains* involves finding their stationary state probabilities. To do so, we consider the equation:

$$\boldsymbol{\pi} = \boldsymbol{\pi} P \quad \text{or equivalently} \quad \mathbf{0} = \boldsymbol{\pi}(P - I). \tag{11.68a}$$

In general the above set of linear equations is undetermined. To see this, note that the sum of the columns of the matrix $P - I$ is zero. Therefore we need the normalization equation: $\pi_1 + \pi_2 + \cdots + \pi_K = 1$. We can incorporate this equation by replacing one of the columns of $P - I$ with the all 1's column vector. Let $Q$ be the matrix that results when we replace the first column of $P - I$; the system of linear equations becomes:

$$\mathbf{b} = \boldsymbol{\pi} Q, \tag{11.68b}$$

where $\mathbf{b}$ is a row vector with 1 in the first entry and zeros elsewhere. If the Markov chain is irreducible, then a unique stationary pmf exists and is obtained by inverting the above equation.

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**Example 11.46 Google PageRank**

Find the stationary pmf for the PageRank algorithm in Example 11.30.

After we take $P - I$ from the example and replace the first column with all 1's we obtain:

$$Q = \begin{bmatrix} 1 & 0.4550 & 0.4550 & 0.0300 & 0.0300 \\ 1 & -0.8000 & 0.2000 & 0.2000 & 0.2000 \\ 1 & 0.3133 & -0.9700 & 0.3133 & 0.0300 \\ 1 & 0.0300 & 0.0300 & -0.9700 & 0.8800 \\ 1 & 0.0300 & 0.4550 & 0.0300 & -0.5450 \end{bmatrix}.$$

We then invert $Q$ to obtain the pmf:

$$\boldsymbol{\pi} = (0.13175, 0.18772, 0.24642, 0.13172, 0.30239).$$

The Octave commands for the above procedure are given below:

```
> Q=[1 0.455 0.455 0.03 0.03
> 1 -.8 .2 .2 .2
> 1 0.3133 -.97 0.3133 0.03
> 1 0.03 0.03 -0.97 0.88
```

```
> 1 0.03 0.455 0.03 −.545];
> b=[1 0 0 0 0];
> p=b*inv(Q)
p =
  0.13175 0.18772 0.24642 0.13172 0.30239
```

In the case of infinite-state Markov chains, we can apply matrix inversion by truncating the state space at some value where the state probabilities become negligible. Another method, discussed in the next chapter, involves the application of the probability generating function for the state of the system.

To find the stationary pmf for *finite-state continuous-time Markov chains*, we need to find a pmf that satisfies Eq. (11.40a) as well as the normalization condition:

$$\mathbf{0} = \mathbf{p}\Gamma \quad \text{and} \quad 1 = \mathbf{pe} \tag{11.69a}$$

where

$$\Gamma = \begin{bmatrix} -v_0 & \gamma_{01} & \gamma_{02} & \gamma_{03} \\ \gamma_{10} & -v_1 & \cdots & \gamma_{1K-1} \\ \cdots & \cdots & \cdots & \cdots \\ \gamma_{K-1\,0} & \gamma_{K-1\,1} & \cdots & -v_{K-1} \end{bmatrix} \quad \text{and} \quad \mathbf{e} = \begin{bmatrix} 1 \\ 1 \\ \cdots \\ 1 \end{bmatrix}. \tag{11.69b}$$

The columns of $\Gamma$ sum to zero, so as before we need to replace a column of $\Gamma$ with $\mathbf{e}$. We obtain $\mathbf{p}$ by multiplying $\mathbf{b}$ by the inverse of the resulting matrix.

**Example 11.47 Cartridge Inventory**

An office orders laser printer cartridges in batches of four cartridges. Suppose that each cartridge lasts for an exponentially distributed time with mean 1 month. Assume that a new batch of four cartridges becomes available as soon as the last cartridge in a batch runs out. Find the stationary pmf for $N(t)$, the number of cartridges available at time $t$.

$N(t)$ takes on values from the set $\{1, 2, 3, 4\}$ and follows a periodic sequence of values $4 \to 3 \to 2 \to 1 \to 4 \ldots$. The rate out of each state is 1 and the rate into each state from the previous state is also 1. Therefore the transition rate matrix and the modified global balance equations are:

$$\Gamma = \begin{bmatrix} -1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \qquad \mathbf{b} = \mathbf{p}\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & 0 & 1 & -1 \end{bmatrix}.$$

It is easy to show that the $\mathbf{p} = (1/4, 1/4, 1/4, 1/4)$. In a more complicated case we would use numerical inversion to solve for $\mathbf{p}$.

### 11.6.2 Time-Dependent Probabilities of Markov Chains

We now consider finding the time-dependent probabilities of a *finite-state discrete-time Markov chain* as given by Eq. (8.16b). Example 11.9 described the general approach for finding $P^n$. First, however, we note a few facts about the transition probability matrix $P$.

A **stochastic matrix** is defined as a nonnegative matrix for which the elements of each row add to one. Thus $P$ is a stochastic matrix. A stochastic matrix always has $\lambda = 1$ as an eigenvalue and $\mathbf{e}^T = (1, \ldots, 1)$ as a right eigenvector: $1\mathbf{e} = P\mathbf{e}$. This follows from the fact that all the row elements of $P$ add to one. On the other hand, the stationary pmf $\boldsymbol{\pi}$ is a left eigenvector for the $\lambda = 1$ eigenvalue of $P$: $1\boldsymbol{\pi} = \boldsymbol{\pi}P$. It can be shown [Gallager, pp. 116–117] that if $P$ corresponds to an aperiodic irreducible Markov chain, then $\lambda = 1$ is the largest eigenvalue and the magnitude of all other eigenvalues are less than 1.

Let $P$ correspond to an aperiodic irreducible Markov chain. Proceeding as in Example 11.19, to find $P^n$ we first find the eigenvalues $1 = \lambda_1 > |\lambda_2| > \ldots > |\lambda_K|$ and right eigenvectors of $P$: $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_K$. Letting $\mathbf{E}$ be the matrix with eigenvectors as columns, we then have that:

$$\begin{aligned} P^n &= \mathbf{E}\Lambda^n\mathbf{E}^{-1} \\ &= \mathbf{E}\begin{bmatrix} 1 & 0 & \ldots & 0 \\ 0 & \lambda_2^n & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & \lambda_K^n \end{bmatrix}\mathbf{E}^{-1}. \end{aligned} \tag{11.70}$$

Note how all but the 1-1 entry in the diagonal matrix approach zero as $n$ increases. Note as well that the first column of $\mathbf{E}$ is the all 1's vector. This implies that the first row of $\mathbf{E}^{-1}$ contains the stationary pmf $\boldsymbol{\pi}$. In Octave the eigenvalues and eigenvectors of $P$ are obtained using the `eig(P)` function, which was discussed previously in Section 10.7. In practice it is simpler and more convenient to use the command `P^n`.

Next we consider finding the time-dependent probabilities of a *finite-state continuous-time Markov chain* that are the solution to Eq. (11.39):

$$\mathbf{p}'(t) = [p_j'(t)] = \sum_{i=1}^{K} p_i(t)\gamma_{ij} = \mathbf{p}(t)\Gamma \quad \text{subject to} \quad \mathbf{p}(0) = (p_i(0), \ldots, p_K(0)). \tag{11.71}$$

We are now dealing with first-order vector differential equations. Electrical engineering students encounter this equation in an introductory linear systems course. The solution is given by:

$$\mathbf{p}(t) = \mathbf{p}(0)P(t) = \mathbf{p}(0)e^{\Gamma t} \tag{11.72a}$$

where $P(t) = e^{\Gamma t}$ is the matrix of transition probabilities in an interval of length $t$ seconds, and where the exponential matrix function is defined by:

$$P(t) = [p_{ij}(t)] = e^{\Gamma t} = \sum_{j=0}^{\infty} \frac{(\Gamma t)^j}{j!}. \tag{11.72b}$$

Furthermore, using matrix diagonalization the exponential matrix can be evaluated as:

$$P(t) = \mathbf{E}[e^{\lambda_i t}]\mathbf{E}^{-1} \tag{11.72c}$$

where $\mathbf{E}$ is a matrix whose columns are the eigenvectors of $\Gamma$ and the middle matrix is a diagonal matrix with exponential functions as its elements. [Gallager, p. 194] shows

that if the Markov chain is finite state and irreducible, then $\Gamma$ has an eigenvalue $\lambda = 0$ which has right eigenvector $\mathbf{e}^T = (1, 1,...,1)$. $\Gamma$ also has a left eigenvector $\mathbf{p}$ corresponding to $\lambda = 0$ which is the unique stationary state pmf. Furthermore the remaining eigenvalues of $\Gamma$ have negative real parts. This implies that all but the $\lambda = 0$ exponential terms in the diagonal matrix decay to zero as $t$ increases. If we let $\lambda = 0$ occupy the 1-1 entry in the diagonal matrix, then as $t \to \infty$, $P(t)$ approaches the product of the $\mathbf{e}$ and the first row of $\mathbf{E}^{-1}$.

### Example 11.48 Cartridge Inventory

Find the state probabilities for $N(t)$ in Example 11.47 if $N(0) = 4$.

We use the `eig(Γ)` function to obtain the eigenvalues and eigenvectors of $\Gamma$ and the associated matrices, $\mathbf{E}$, $\Lambda$, and $\mathbf{E}^{-1}$:

$$\mathbf{E} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \mathrm{j} & -j & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -j & j & -1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1-j & 0 & 0 \\ 0 & 0 & -1+j & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} \quad \mathbf{E}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{j} & -1 & j \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

Note that two of the eigenvalues and their corresponding eigenvectors are complex. The state probabilities are given by:

$$\mathbf{p}(t) = \mathbf{p}(0)\mathbf{E}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-(1+j)t} & 0 & 0 \\ 0 & 0 & e^{-(1-j)t} & 0 \\ 0 & 0 & 0 & e^{-2t} \end{bmatrix}\mathbf{E}^{-1}$$

$$= (0,0,0,1)\frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -j & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -j & j & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-(1+j)t} & 0 & 0 \\ 0 & 0 & e^{-(1-j)t} & 0 \\ 0 & 0 & 0 & e^{-2t} \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$= \frac{1}{4}(1,-j,j,-1)\begin{bmatrix} 1 & 1 & 1 & 1 \\ e^{-(1+j)t} & -je^{-(1+j)t} & -e^{-(1+j)t} & je^{-(1+j)t} \\ e^{-(1-j)t} & je^{-(1-j)t} & -e^{-(1-j)t} & -je^{-(1-j)t} \\ e^{-2t} & -e^{-2t} & e^{-2t} & -e^{-2t} \end{bmatrix}$$

$$= \frac{1}{4}(1 - 2e^{-t}\sin t - e^{-2t}, 1 - 2e^{-t}\cos t + e^{-2t}, 1 + 2e^{-t}\sin t - e^{-2t}, 1 + 2e^{-t}\cos t + e^{-2t}).$$

Figure 11.18 shows the four state probabilities vs. time. It can be seen that all of the probability mass is initially in state 4 and that the mass first transfers to state 3, then state 2, and finally to state 1. Eventually all state probabilities approach the steady state value of 1/4.

## 11.6.3 Simulation of Markov Chains

We simulate a Markov chain by emulating its underlying random experiments. We begin by selecting the initial state according to an initial state pmf. We then generate the sequence of states by producing outcomes according to the associated transition

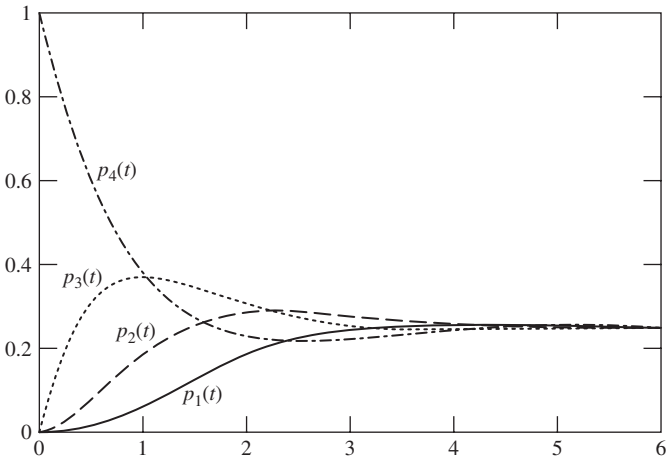



**FIGURE 11.18**
Time-dependent probabilities in cartridge inventory.

probabilities. In the case of continuous-time Markov chains we also need to generate a state occupancy time after each state transition has been determined. Figure 11.19 shows the inputs and outputs of generic modules for generating realizations of a Markov chain.

**Discrete-Time Markov Chains** The module for generating a sequence of states for a Markov chain requires the following inputs: i. The state space; ii. The matrix of state transition probabilities; iii. The initial state probability mass function; and iv. The number of steps in the simulation sequence. The module operates as follows:

1. Generates the initial state according to $\pi_0$.
2. Repeatedly generates the next state according to the transition probabilities of the current state.
3. Stops when the required number of steps has been simulated.

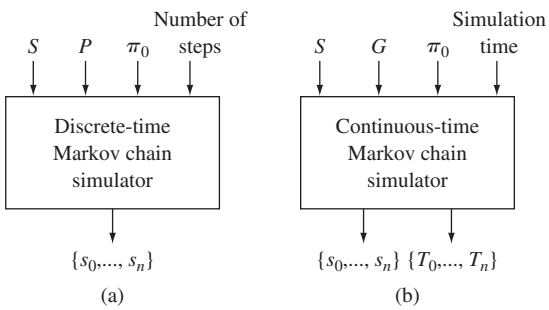



**FIGURE 11.19**
Generic modules for simulating Markov chains.

### Example 11.49 Discrete-Time Markov Chain

Develop a program to generate Markov chains with the state transition diagram as shown in Fig. 11.20(a). Note that the Markov chain is similar to that of a birth-death process except that transitions from a state to itself are allowed. Use the program to simulate 1000 time steps in a data multiplexer where in each time unit a packet is received with probability $a$, and/or a packet transmitted from its buffer with probability $b$. Assume the data multiplexer is initially empty.

For this example we wrote the function `Discrete_MC(Nmax,P,IC,L)`. The state space is $\{0, 1, \ldots, N_{max}\}$. Since Octave uses indices from 1 onwards, the array state ranges from 1 to $N_{max} + 1$. For the Markov chains under consideration we need to specify only three probabilities for the transition probabilities for each state. Therefore $P$ is an $N_{max} + 1$ row by 3 column matrix. The initial state pmf is a $N_{max} + 1$ by 1 vector. The output of the function is a vector of states of size $L$.

The Markov chain for the data multiplexer has the following transition probabilities. If $N = 0$, that is, the system is empty, the next state is either $N = 1$ with probability $a$, or $N = 0$ with probability $1 - a$, that is: $p_{00} = 1 - a$, $p_{01} = a$. If $N = n > 0$, the next state is $n + 1$ with probability $(1 - b)a$; $n$ with probability $ab$; or $n - 1$ with probability $b(1 - a)$, that is: $p_{n\,n+1} = (1 - b)a$, $p_{n\,n} = ab$, $p_{n-1\,n} = (1 - a)b$. If $N = N_{max}$, the next state is $N_{max} - 1$ with probability $(1 - a)b$; or $N_{max}$ with probability $1 - b(1 - a)$, since the system is not allowed to grow beyond $N_{max}$.

The code below prepares the inputs and then calls the function `Discrete_MC(S, P,IC,N)`. The basic step in the function involves generating a discrete random variable that determines whether the chain increases by 1, decreases by 1, or remains the same.

```
Nmax=50;
P=zeros(Nmax+1,3);
a=0.45;
b=0.50;
P(1,:)=[0,1-a,a];
r=[(1-a)*b,a*b+(1-a)*(1-b),(1-b)*a];
for n=2:Nmax;
    P(n,:)=r;
end
```

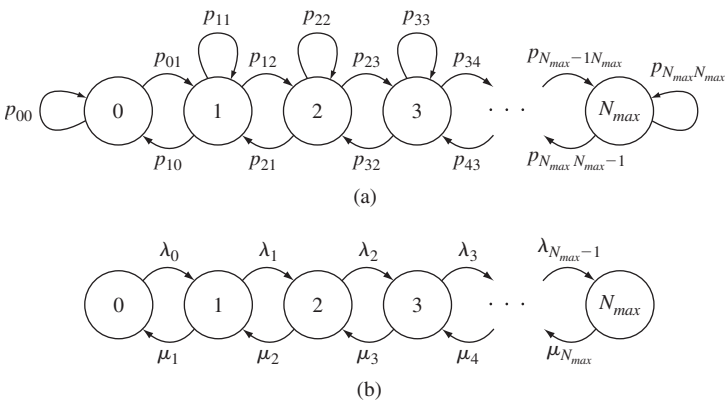



**FIGURE 11.20**
Generic Markov chains: (a) discrete-time; (b) birth-death continuous-time.